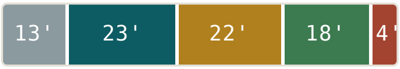
Home / Grade 7 / Geometry / Quarter 1

The simplest geometric figures

Point, line and plane — the three words that are never defined, and everything built from them.

LESSON 1–2 | 2 × 40 MIN | MATEMATIKA 7, §1 | S8 5

LESSON CLOCK



Three undefined words	13 min
Notation	23 min
The axioms	22 min
Counting lines	18 min
Homework	4 min

LEARNING OBJECTIVES

- Name the point, the line and the plane as the undefined basic objects.
- Use the notation for points, lines, rays and segments correctly.
- State the axiom that two points determine exactly one line.
- Decide how many lines pass through a given set of points.

TEXTBOOK

National	pp. 3–8
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01 Explanation

Three undefined words

Geometry begins with three ideas that are **not** defined: the **point**, the **line** and the **plane**. Everything else is defined in terms of them.

OBJECT	WRITTEN	DRAWN AS
point	A, B, C	a dot
line	a, b or AB	a stroke with no ends
plane	α, β	a parallelogram

WHY NOT DEFINE THEM?

Every definition uses simpler words. Somewhere the chain has to stop, and mathematics chooses to stop at these three. Instead of definitions they are given **axioms** — statements accepted without proof that say how they behave.

Notation

FIGURE	WRITTEN	HAS
line AB	AB	no ends — it continues both ways
ray AB	$[AB)$	one end, at A
segment AB	$[AB]$	two ends
A lies on a	$A \in a$	—
A does not lie on a	$A \notin a$	—

A **ray** is half a line: it starts at a point and runs on forever in one direction. Two rays $[AB)$ and $[BA)$ on the same line point in opposite directions.

> A LINE HAS NO ENDS AND NO LENGTH
>
> Only a segment has a length. Asking for "the length of the line AB" is asking for something that does not exist — the phrase must be "the length of the segment AB".

The axioms

AXIOM	STATEMENT
1	through any two points there passes exactly one line
2	two different lines meet in at most one point
3	on every line there are at least two points
4	a line divides the plane into two half-planes

Axiom 1 is the one used constantly. It is why a ruler needs only two marks to draw a line, and why two nails hold a plank rigid.

> AXIOM 2 FOLLOWS FROM AXIOM 1
>
> If two lines met in two points, then two points would have two lines through them — which Axiom 1 forbids. Deriving one statement from another is the whole activity of geometry, and it starts here.

Counting lines

How many lines pass through n points, no three of them on one line?

N	LINES	WHY
2	1	Axiom 1
3	3	each pair
4	6	$\frac{4 \times 3}{2}$
5	10	$\frac{5 \times 4}{2}$
n	$\frac{n(n-1)}{2}$	each pair, counted once

“NO THREE ON ONE LINE” IS ESSENTIAL

If all n points lie on one line, there is exactly **one** line, not $\frac{n(n-1)}{2}$. The condition is what makes the counting work.

02 Worked examples

EXAMPLE 1 **How many lines pass through 5 points, no three on one line?**

1. Each pair of points gives one line.
   Axiom 1.
2. Pairs from 5: $\frac{5 \times 4}{2}$.
   Each counted once.
3. $= 10$
4. All ten are different, since no three points are collinear.

ANSWER

10

EXAMPLE 2 **Four points lie on a plane with exactly three of them on one line. How many lines are there?**

1. The three collinear points give 1 line.
   Not 3.

2. The fourth point joins each of the three: 3 more.

3. $1 + 3$

4. $= 4$ lines.

ANSWER

4

EXAMPLE 3 **Name all the rays and segments determined by three points A, B, C on one line, in that order.**

1. Segments: $[AB]$, $[BC]$, $[AC]$.
   Three.

2. Rays from A: $[AB)$ — one direction only.

3. From B: $[BA)$ and $[BC)$.

4. From C: $[CB)$. Four rays in all.

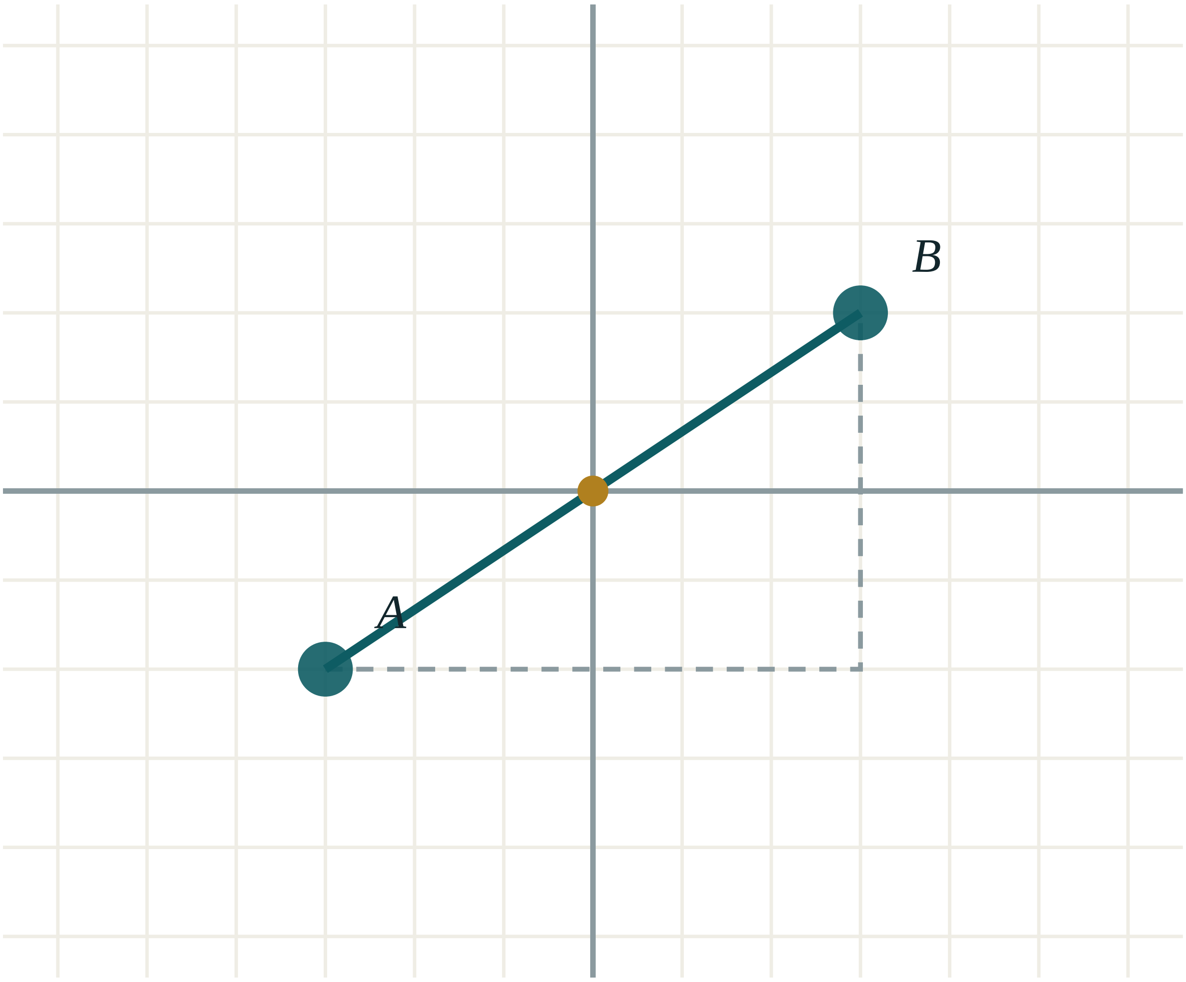
ANSWER

3 segments, 4 rays

03 Interactive model

Stretch a thread between two drawing pins and ask whether a second thread could take a different route; Axiom 1 is demonstrated before it is stated.

Points and the lines through them



A **(-3, -2)** | B **(3, 2)** | $x_2 - x_1$ **6** | $y_2 - y_1$ **4** | distance AB **7.21** | midpoint M **(0, 0)**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Point	Nuqta	Точка
Line	To‘g‘ri chiziq	Прямая
Plane	Tekislik	Плоскость
Ray	Nur	Луч
Segment	Kesma	Отрезок
Axiom	Aksioma	Аксиома
To belong to	Tegishli bo‘lmoq	Принадлежать
Intersection	Kesishish	Пересечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which are undefined in geometry?

- segment, ray, angle
- point, line, plane
- triangle, circle, square
- none

Through two points there passes:

- no line
- exactly one line
- two lines
- many lines

A ray has:

- no ends
- one end
- two ends
- a length

Which has a length?

- a line
- a ray
- a segment
- a plane

5 points, no three collinear: how many lines?

- 5
- 10
- 20
- 25

Two different lines meet in:

- no point
- at most one point
- two points
- many points

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. How many lines through 2 points?

   ANSWER 1

2. How many through 3 points, no three collinear?

   ANSWER 3

3. How many through 4?

   ANSWER 6

4. How many through 5?

   ANSWER 10

5. Which figure has two ends?

   ANSWER A segment

6. Which has one end?

   ANSWER A ray

7. Which has none?

   ANSWER A line

Medium · the standard question

7 PROBLEMS

1. How many through 6 points, no three collinear?

   ANSWER 15

2. How many through 10?

   ANSWER 45

3. Four points, three of them collinear

   ANSWER 4 lines

4. Five points all on one line

   ANSWER 1 line

5. Segments determined by 4 points on a line

   ANSWER 6

6. Rays determined by 3 points on a line

   ANSWER 4

7. Two lines meet in at most

   ANSWER One point

Hard · stretch and reason

7 PROBLEMS

1. For which n do n points give 28 lines?

   ANSWER $n = 8$

2. Five points with exactly four collinear

   ANSWER 5 lines

3. Six points, three collinear and three collinear on another line

   ANSWER 11 lines

4. How many segments have n points on a line as ends?

   ANSWER $\frac{n(n-1)}{2}$

5. How many rays?

   ANSWER $2(n-1)$

6. Why can two lines not meet twice?

   ANSWER Axiom 1 would be broken

7. A plane is divided by 3 lines into at most

   ANSWER 7 parts

07 Homework

Homework — 5 tasks

Draw a figure for every counting question; the picture is the proof.

1. How many lines pass through 7 points, no three of them collinear?
2. Four points lie in a plane, with exactly three on one line. How many lines do they determine?
3. Name all the segments determined by four points on one line.
4. State the four axioms of this lesson in your own words.

5. Explain why a line has no length but a segment does.

The segment: comparing and measuring

Two ends, one number — and the one property that makes measurement possible.

LESSON 3–4 | 2 × 40 MIN | MATEMATIKA 7, §2 | S8 5

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Comparing	13 min
Measuring	23 min
Adding segments	22 min
The midpoint	18 min
Homework	4 min

LEARNING OBJECTIVES

- Compare two segments by superposition and by measurement.
- Use the additive property AB = AC + CB when C lies between A and B.
- Find the midpoint of a segment and use it.
- Convert between the units of length and solve simple problems.

TEXTBOOK

National	pp. 9–14
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01 Explanation

Comparing

Two segments are **equal** if one can be laid exactly on the other. In practice they are compared with compasses: open the compasses to AB and see whether the opening reaches from C to D.

RESULT	WRITTEN
the opening fits exactly	$AB = CD$
the opening falls short	$AB < CD$
the opening overshoots	$AB > CD$

COMPARISON COMES BEFORE MEASUREMENT

Two segments can be compared without any ruler at all. Measurement is comparison with one chosen segment — the unit — repeated. That is the whole idea of a length.

Measuring

Choose a **unit segment**. The length of AB is the number of units it contains.

UNIT	SYMBOL	IN METRES
millimetre	*mm*	*0.001*
centimetre	*cm*	*0.01*
decimetre	*dm*	*0.1*
metre	*m*	*1*
kilometre	*km*	*1000*

Each step up the table is a factor of 10, except metre to kilometre, which is 1000.

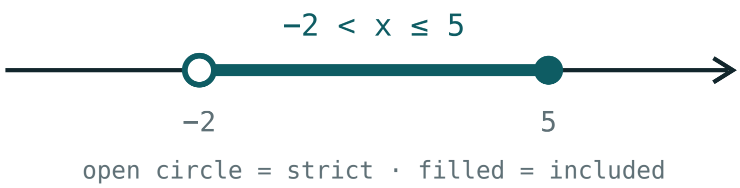
A LENGTH IS A NUMBER *AND* A UNIT

“The segment is 7” means nothing until the unit is said. Every answer in this chapter carries its unit, and a missing unit costs a mark.

Adding segments

If C lies **between** A and B, then

$$AB = AC + CB$$



A point between the ends splits the segment into two, and the lengths add.

AC	CB	AB
3 cm	5 cm	8 cm
4.5 cm	2.5 cm	7 cm
12 mm	?	30 mm $\Rightarrow$ 18 mm

“BETWEEN” IS A CONDITION, NOT A DECORATION

If C is not between A and B the equation is false. With A, B, C in that order, $AC = AB + BC$ instead — so the order of the points must be established from the figure first.

The midpoint

The **midpoint** M of AB is the point between A and B with $AM = MB$. Then

$$AM = MB = \frac{\mathrm{AB}}{2}$$

GIVEN	FIND	WORKING
$AB = 14$ cm	AM	7 cm
$AM = 6.5$ cm	AB	13 cm
M and N midpoints of AC and CB	MN	$\frac{\mathrm{AB}}{2}$

The third row is a small surprise worth checking: whatever the position of C between A and B, the distance between the two midpoints is always half of AB.

A SEGMENT HAS EXACTLY ONE MIDPOINT

Not two, and never none. The construction with compasses — equal arcs from both ends — produces it exactly, and it is the first construction of the course.

02 Worked examples

EXAMPLE 1 **C lies between A and B. If $AC = 12$ mm and $AB = 30$ mm, find CB.**

1. $AB = AC + CB$
2. $30 = 12 + CB$
3. $CB = 18$ mm
4. Check: $12 + 18 = 30$ ✓

ANSWER

18 mm

EXAMPLE 2 **$AB = 14$ cm and M is its midpoint. N is the midpoint of AM. Find NB.**

1. $AM = 7$ cm
2. $AN = 3.5$ cm
3. $NB = AB - AN = 14 - 3.5$
4. $= 10.5$ cm

ANSWER

10.5 cm

EXAMPLE 3 **Convert *2.4 m* to centimetres and *35 mm* to metres.**

1. $1\ m = 100\ cm$
2. $2.4 \times 100 = 240\ cm$
3. $1\ mm = 0.001\ m$
4. $35 \times 0.001 = 0.035\ m$

ANSWER

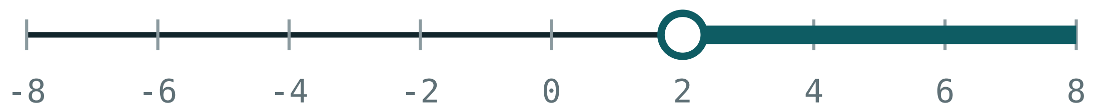
$240\ cm$ and $0.035\ m$

03 Interactive model

Compare two segments drawn on the board with compasses only, then measure both; the class sees that comparison needs no numbers and measurement needs a unit.

Points on a segment

$$x > 2$$



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Segment	Kesma	Отрезок
Length	Uzunlik	Длина
Midpoint	O‘rta nuqta	Середина
Between	Orasida	Между
To superpose	Ustma-ust qo‘ymoq	Наложить
Equal segments	Teng kesmalar	Равные отрезки
Unit of length	Uzunlik birligi	Единица длины
Additive property	Additivlik xossasi	Свойство аддитивности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Two segments are equal when:

- they look alike
- one lays exactly on the other
- they are near
- they are drawn together

$AB = AC + CB$ requires:

- C anywhere
- C between A and B
- C outside
- nothing

$AM = MB$ makes M:

- an end
- the midpoint
- outside
- a ray

$AB = 14\ cm$: $AM =$

- $7\ cm$
- $14\ cm$
- $28\ cm$
- $3.5\ cm$

$1\ m$ equals:

- $10\ cm$
- $100\ cm$
- $1000\ cm$
- $10\ mm$

A length must have:

- a number only
- a unit only
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $AC = 3, CB = 5$: AB

ANSWER 8

2. $AC = 12, AB = 30$: CB

ANSWER 18

3. $AB = 14$: AM

ANSWER 7

4. $AM = 6.5$: AB

ANSWER 13

5. $2.4\ m$ in cm

ANSWER 240

6. $35\ mm$ in m

ANSWER 0.035

7. $3\ km$ in m

ANSWER 3000

Medium · the standard question

7 PROBLEMS

1. $AB = 14$, M midpoint, N midpoint of AM: NB

   ANSWER 10.5

2. $AC = 4.5$, $CB = 2.5$: AB

   ANSWER 7

3. $AB = 25$, $AC = 9$: CB

   ANSWER 16

4. $850\ cm$ in m

   ANSWER 8.5

5. $0.6\ km$ in cm

   ANSWER $60\ 000$

6. A segment of $18\ cm$ divided in $2 : 1$

   ANSWER 12 and 6

7. Midpoints of AC and CB with $AB = 20$: MN

   ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. $AB = 24$, C with $AC : CB = 3 : 5$: AC

   ANSWER 9

2. Three points A, B, C with $AB = 5$, $BC = 3$: possible AC

   ANSWER 8 or 2

3. M is the midpoint of AB and $AM = 3x - 1$, $MB = x + 7$: AB

   ANSWER 22

4. A segment is divided into 4 equal parts; the distance between the 1st and 3rd division points, if $AB = 32$

   ANSWER 16

5. $AB = 30$ with C between; AC exceeds CB by 6: AC

   ANSWER 18

6. A ruler of $30\ cm$ measures a table 7 times exactly: the table

   ANSWER $2.1\ m$

7. Why is the midpoint unique?

   ANSWER Two midpoints would give two different halves

07 Homework

Homework — 5 tasks

Every answer carries a unit, and every figure carries its labels.

1. C lies between A and B with $AC = 17\ mm$ and $AB = 42\ mm$. Find CB.
2. $AB = 18\ cm$ and M is its midpoint. Find AM and MB.
3. Convert $4.7\ m$ to millimetres and $2500\ cm$ to kilometres.
4. A segment of $45\ cm$ is divided in the ratio $2 : 3$. Find the two parts.

5. Three points lie on a line with $AB = 7$ and $BC = 4$. Give both possible values of AC.

The angle: comparing and measuring

Two rays from one point — and the protractor that turns them into a number.

LESSON 5–6 | 2 × 40 MIN | MATEMATIKA 7, §3 | S8 5

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

What an angle is	13 min
Measuring	23 min
The five kinds	22 min
Adding angles	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define an angle, its vertex and its arms, and use the three-letter notation.
- Measure and draw angles with a protractor.
- Classify angles as acute, right, obtuse, straight or reflex.
- Use the additive property and the bisector of an angle.

TEXTBOOK

National	pp. 15–21
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01 Explanation

What an angle is

An **angle** is the figure formed by two rays with a common origin. The origin is the **vertex** and the rays are the **arms**.

NOTATION	MEANS
$\angle ABC$	vertex B, arms through A and C
$\angle B$	the same, when no confusion is possible
$\angle 1$, $\angle \alpha$	a label written inside the angle

THE VERTEX LETTER GOES IN THE MIDDLE

$\angle ABC$ and $\angle CBA$ are the same angle; $\angle BAC$ is a different one. Reading the middle letter first is the habit to build.

Measuring

The unit is the **degree**: a full turn is divided into 360 equal parts. A protractor is a half-turn of 180 of them.

STEP	WHAT TO DO
1	put the centre of the protractor on the vertex
2	lay the zero line along one arm
3	read where the other arm crosses the scale
4	choose the scale that starts at 0 on your arm

Smaller units exist: $1° = 60'$ (minutes) and $1' = 60''$ (seconds), used in astronomy and surveying.

A PROTRACTOR HAS TWO SCALES

One runs 0 to 180 left to right, the other right to left. Reading $130°$ when the angle is plainly acute means the wrong scale was used — and a glance at the figure catches it.

The five kinds

NAME	SIZE	EXAMPLE
acute	$0° < \alpha < 90°$	$45°$
right	$\alpha = 90°$	a corner of a page
obtuse	$90° < \alpha < 180°$	$130°$
straight	$\alpha = 180°$	a line through the vertex
reflex	$180° < \alpha < 360°$	$250°$

Two angles are **complementary** if they add to $90°$ and **supplementary** if they add to $180°$.

COMPLEMENTARY IS THE SMALLER PAIR

C comes before S in the alphabet, and 90 before 180. That is the whole mnemonic, and it prevents the one confusion this vocabulary causes.

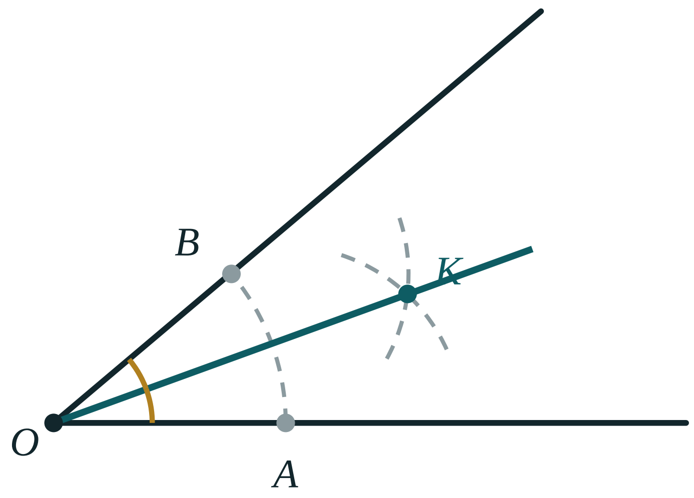
Adding angles

If the ray *[BD)* lies inside $\angle ABC$, then

$$\angle ABC = \angle ABD + \angle DBC$$

The **bisector** of an angle is the ray from the vertex that divides it into two equal angles:

$$\angle ABD = \angle DBC = \frac{\angle \mathrm{ABC}}{2}$$



The bisector divides the angle into two equal parts.

GIVEN	FIND	WORKING
$\angle ABC = 80°$	half	$40°$
$\angle ABD = 35°, \angle DBC = 48°$	$\angle ABC$	$83°$
$\angle ABC = 140°, \angle ABD = 55°$	$\angle DBC$	$85°$

THE SAME ADDITIVE LAW AS FOR SEGMENTS

A point between the ends splits a segment; a ray inside the angle splits the angle. One idea, two settings — which is why the two lessons sit side by side.

02 Worked examples

EXAMPLE 1 **$\angle ABD = 35°$ and $\angle DBC = 48°$, with $[BD)$ inside $\angle ABC$. Find $\angle ABC$.**

1. $\angle ABC = \angle ABD + \angle DBC$
2. $= 35° + 48°$
3. $= 83°$
4. Acute, since $83° < 90°$.

ANSWER

$83°$

EXAMPLE 2 **The bisector of $\angle ABC = 140°$ is $[BD)$. Find $\angle ABD$, and classify both angles.**

1. $\angle ABD = \dfrac{140°}{2}$
2. $= 70°$
3. $\angle ABC = 140°$ is obtuse.
4. $\angle ABD = 70°$ is acute.

ANSWER

$70°$; obtuse and acute

EXAMPLE 3 **Find the complement and the supplement of $37°$.**

1. Complement: $90° - 37°$.
2. $= 53°$
3. Supplement: $180° - 37°$.
4. $= 143°$

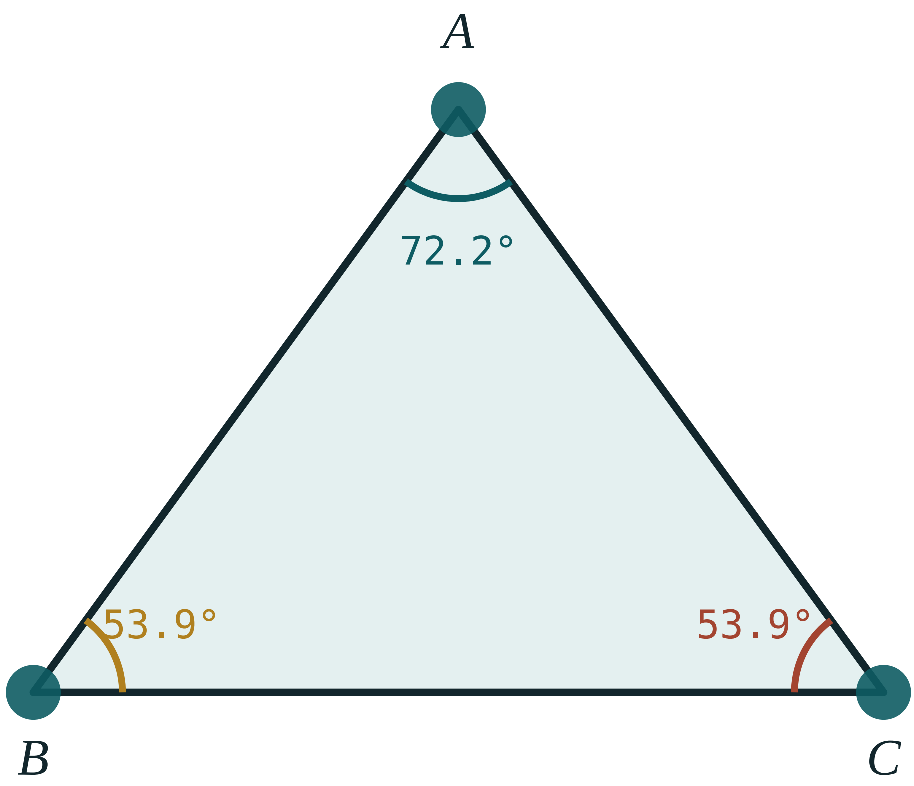
ANSWER

$53°$ and $143°$

03 Interactive model

Open a pair of scissors slowly and ask the class to call out “acute”, “right”, “obtuse”, “straight”; the classification becomes a reflex before it is a table.

Measuring and classifying



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Arm of an angle	Burchak tomoni	Сторона угла
Protractor	Transportir	Транспортир
Acute angle	O‘tkir burchak	Острый угол
Obtuse angle	O‘tmas burchak	Тупой угол
Straight angle	Yoyiq burchak	Развёрнутый угол
Bisector	Bissektrisa	Биссектриса

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In $\angle ABC$ the vertex is:

- A
- B
- C
- all three

A right angle is:

- $45°$
- $90°$
- $180°$
- $360°$

$130°$ is:

- acute
- right
- obtuse
- reflex

Complementary angles add to:

- $90°$
- $180°$
- $270°$
- $360°$

The supplement of $37°$:

- $53°$
- $143°$
- $323°$
- $37°$

A bisector divides an angle into:

- two equal parts
- three parts
- two unequal parts
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Classify $45°$

ANSWER Acute

2. Classify $90°$

ANSWER Right

3. Classify $130°$

ANSWER Obtuse

4. Classify $180°$

ANSWER Straight

5. Classify $250°$

ANSWER Reflex

6. Complement of $37°$

ANSWER $53°$

7. Supplement of $37°$

ANSWER $143°$

Medium · the standard question

7 PROBLEMS

1. $\angle ABD = 35°$, $\angle DBC = 48°$: $\angle ABC$

   ANSWER $83°$

2. $\angle ABC = 140°$, $\angle ABD = 55°$: $\angle DBC$

   ANSWER $85°$

3. Bisector of $140°$

   ANSWER $70°$ each

4. Bisector of $96°$

   ANSWER $48°$ each

5. Complement of $64°$

   ANSWER $26°$

6. Supplement of $118°$

   ANSWER $62°$

7. An angle equal to its own supplement

   ANSWER $90°$

Hard · stretch and reason

7 PROBLEMS

1. An angle is 3 times its complement

   ANSWER $67.5°$

2. An angle is 4 times its supplement

   ANSWER $144°$

3. An angle exceeds its complement by $20°$

   ANSWER $55°$

4. Two supplementary angles in the ratio $4 : 5$

   ANSWER $80°$ and $100°$

5. The angle between the bisectors of two adjacent supplementary angles

   ANSWER $90°$

6. The angle between the hands of a clock at $3{:}30$

   ANSWER $75°$

7. At $9{:}20$

   ANSWER $160°$

07 Homework

Homework — 5 tasks

Measure every drawn angle with a protractor and write the reading beside it.

1. Classify $28°$, $90°$, $155°$ and $300°$.
2. Find the complement and the supplement of $54°$.
3. $[BD)$ lies inside $\angle ABC = 118°$ with $\angle ABD = 47°$. Find $\angle DBC$.
4. An angle is twice its complement. Find it.
5. Find the angle between the hands of a clock at $4{:}00$ and at $7{:}30$.

Practical exercises — multiplying and dividing by 0.1 and 0.01, and rounding

A Cambridge insert: the decimal arithmetic that every later measurement depends on.

LESSON 7–8 | 2 × 40 MIN | MATEMATIKA 7, AMALIY MASHQLAR | S8 3.1–3.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Moving the point	13 min
Dividing by a small number	23 min
Rounding	22 min
Estimating	18 min
Homework	4 min

LEARNING OBJECTIVES

- Multiply and divide by 0.1, 0.01 and 0.001 by moving the decimal point.
- Explain why dividing by 0.1 makes a number larger.
- Round to a given number of decimal places and of significant figures.
- Estimate an answer before computing it.

TEXTBOOK

National	pp. 22–25
Cambridge	Stage 8, pp. 26–34
Workbook	Exercise 3.1–3.2

01 Explanation

Moving the point

OPERATION	EFFECT	EXAMPLE
$\times 10$	point moves 1 right	$3.7 \to 37$
$\times 0.1$	point moves 1 left	$3.7 \to 0.37$
$\times 0.01$	point moves 2 left	$3.7 \to 0.037$
$\div 0.1$	point moves 1 right	$3.7 \to 37$
$\div 0.01$	point moves 2 right	$3.7 \to 370$

Multiplying by 0.1 is the same as dividing by 10; dividing by 0.1 is the same as multiplying by 10.

$0.1 = \frac{1}{10}$, AND THAT EXPLAINS EVERYTHING

Dividing by $\frac{1}{10}$ asks "how many tenths are there in it?" — and there are ten times as many tenths as there are wholes.

Dividing by a small number

The result of dividing by a number less than 1 is **larger** than what you started with.
This surprises almost everyone at first.

QUESTION	IN WORDS	ANSWER
$6 \div 2$	how many 2s in 6?	3
$6 \div 1$	how many 1s?	6
$6 \div 0.5$	how many halves?	12
$6 \div 0.1$	how many tenths?	60
$6 \div 0.01$	how many hundredths?	600

"DIVIDING MAKES IT SMALLER" IS FALSE

It is true only when dividing by a number greater than 1. This is one of the most persistent wrong beliefs in school arithmetic, and reading the division as "how many … in …?" cures it.

Rounding

RULE	STATEMENT
look at	the first digit you are discarding
if it is 0–4	leave the kept digit alone
if it is 5–9	add 1 to the kept digit

Decimal places count from the point; **significant figures** count from the first non-zero digit.

NUMBER	2 D.P.	2 S.F.
3.4567	*3.46*	*3.5*
0.004829	*0.00*	*0.0048*
1954	*1954.00*	*2000*

LEADING ZEROS ARE NOT SIGNIFICANT

In *0.004829* the first significant figure is the *4*. That is why the two columns above differ so sharply for that row.

Estimating

Round every number to *1* significant figure, compute, and compare with the exact answer.

CALCULATION	ESTIMATE	EXACT
4.8×21.3	$5 \times 20 = 100$	*102.24*
$39.6 \div 0.21$	$40 \div 0.2 = 200$	*188.6*
6.9×0.48	$7 \times 0.5 = 3.5$	*3.312*

ESTIMATE FIRST, THEN COMPUTE

An estimate that is ten times out from the answer means a decimal point was misplaced. Thirty seconds of estimating catches errors no amount of re-checking the digits will find.

02 Worked examples

EXAMPLE 1 **Compute 4.7×0.01 and $4.7 \div 0.01$.**

1. $\times\ 0.01$ moves the point 2 places left.
2. $4.7 \times 0.01 = 0.047$
3. $\div\ 0.01$ moves it 2 places right.
4. $4.7 \div 0.01 = 470$

ANSWER

0.047 and 470

EXAMPLE 2 **Round 3.4567 to two decimal places and to two significant figures.**

1. 2 d.p.: keep 3.45, next digit 6.
2. ≈ 3.46
3. 2 s.f.: keep 3.4, next digit 5.
4. ≈ 3.5

ANSWER

3.46 and 3.5

EXAMPLE 3 **Estimate $39.6 \div 0.21$, then say whether 18.86 is a plausible answer.**

1. $40 \div 0.2$

   One significant figure each.

2. $= 200$

3. 18.86 is ten times too small.

4. The true answer is about 189.

   A misplaced point.

ANSWER

No — about 189

03 Interactive model

Ask how many 10-tiyin coins make a sum before doing any division by 0.1; the arithmetic follows from the question, not from a rule.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Decimal point	O‘nli nuqta	Десятичная запятая
Decimal place	O‘nli xona	Десятичный знак
Significant figure	Ma’noli raqam	Значащая цифра
To round	Yaxlitlash	Округлить
Estimate	Baholash	Оценка
Tenth	O‘ndan bir	Десятая
Hundredth	Yuzdan bir	Сотая
Approximation	Taqribiy qiymat	Приближение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\times\ 0.1$ moves the point:

- one right
- one left
- two right
- two left

$\div\ 0.01$ moves the point:

- two left
- two right
- one right
- not at all

$6 \div 0.1$ equals:

- 0.6
- 6
- 60
- 0.06

Dividing by a number less than 1:

- makes it smaller
- makes it larger
- leaves it
- is impossible

3.4567 to 2 s.f.:

- 3.45
- 3.46
- 3.5
- 3.4

In 0.00482 the first significant figure is:

- 0
- 4
- 8
- 2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3.7×0.1

ANSWER 0.37

2. 3.7×0.01

ANSWER 0.037

3. $3.7 \div 0.1$

ANSWER 37

4. $3.7 \div 0.01$

ANSWER 370

5. $6 \div 0.5$

ANSWER 12

6. $6 \div 0.1$

ANSWER 60

7. 4.7×0.01

ANSWER 0.047

Medium · the standard question

7 PROBLEMS

1. 3.4567 to 2 d.p.

   ANSWER 3.46

2. 3.4567 to 2 s.f.

   ANSWER 3.5

3. 0.004829 to 2 s.f.

   ANSWER 0.0048

4. 1954 to 2 s.f.

   ANSWER 2000

5. Estimate 4.8×21.3

   ANSWER 100

6. Estimate $39.6 \div 0.21$

   ANSWER 200

7. $0.42 \div 0.01$

   ANSWER 42

Hard · stretch and reason

7 PROBLEMS

1. $0.6 \times 0.4 \div 0.02$

   ANSWER 12

2. $(0.3)^2 \div 0.09$

   ANSWER 1

3. Estimate $\dfrac{58.7 \times 0.51}{0.19}$

   ANSWER ≈ 150

4. 12.345 to 3 s.f.

   ANSWER 12.3

5. 0.09995 to 3 d.p.

   ANSWER 0.100

6. A rope of $8.4\ m$ cut into $0.35\ m$ pieces

   ANSWER 24 pieces

7. Why does $\div\ 0.1$ multiply by 10?

   ANSWER There are ten tenths in one whole

07 Homework

Homework — 5 tasks

Estimate every calculation to one significant figure before doing it exactly.

1. Compute 5.6×0.1, 5.6×0.001, $5.6 \div 0.1$ and $5.6 \div 0.001$.
2. Round 27.4863 to 1, 2 and 3 decimal places.
3. Round 0.0035729 to 2 and 3 significant figures.
4. Estimate 7.8×0.49 and then compute it exactly.

5. A ribbon of $6.3\ m$ is cut into pieces of $0.45\ m$. How many pieces?

Perpendicular lines

The right angle, made and used — and the shortest distance from a point to a line.

LESSON 9–10 | 2 × 40 MIN | MATEMATIKA 7, §4 | S8 5.3

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Perpendicularity	13 min
Angles at a crossing	23 min
The construction	22 min
The shortest distance	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define perpendicular lines and use the symbol ⊥.
- Construct a perpendicular through a point on a line and off it.
- Know that the perpendicular is the shortest distance from a point to a line.
- Use vertical and adjacent angles at an intersection.

TEXTBOOK

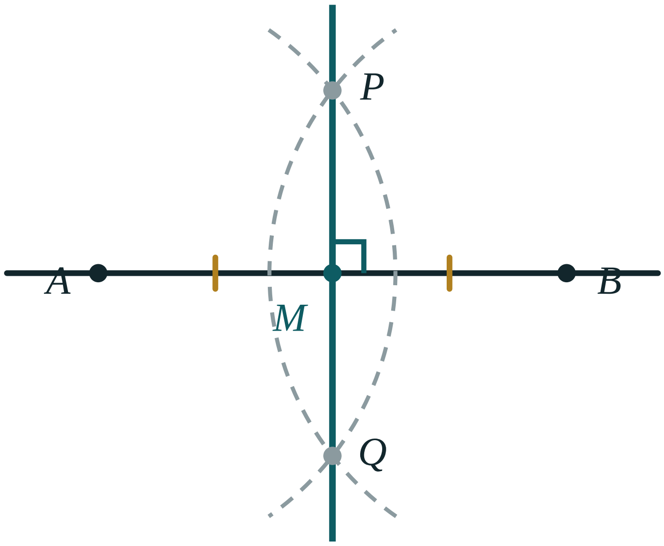
National	pp. 26–31
Cambridge	Stage 8, pp. 53–56
Workbook	Exercise 5.3

01 Explanation

Perpendicularity

Two lines are **perpendicular** if they meet at a right angle. Written $a \perp b$.

When two lines cross, four angles are formed. If one of them is $90°$, all four are — so one right angle at a crossing makes the lines perpendicular.



Perpendicular lines meet at a right angle; the symbol is a small square.

THROUGH A POINT THERE IS EXACTLY ONE PERPENDICULAR TO A LINE

Whether the point is on the line or off it. This uniqueness is what makes "the distance from a point to a line" a well-defined quantity.

Angles at a crossing

PAIR	POSITION	RELATION
adjacent	share an arm, together make a straight line	add to $180°$
vertical	opposite each other	equal

Proof that vertical angles are equal. $\angle 1 + \angle 2 = 180°$ and $\angle 2 + \angle 3 = 180°$, because each pair is adjacent. Subtracting, $\angle 1 = \angle 3$.

GIVEN	THE OTHER THREE
$\angle 1 = 40°$	$140°, 40°, 140°$
$\angle 1 = 90°$	$90°, 90°, 90°$
$\angle 1 = 115°$	$65°, 115°, 65°$

ADJACENT MEANS SHARING AN ARM *AND* MAKING A STRAIGHT LINE

> Two angles that merely touch are not adjacent in this sense. The definition matters, because only genuinely adjacent angles are supplementary.

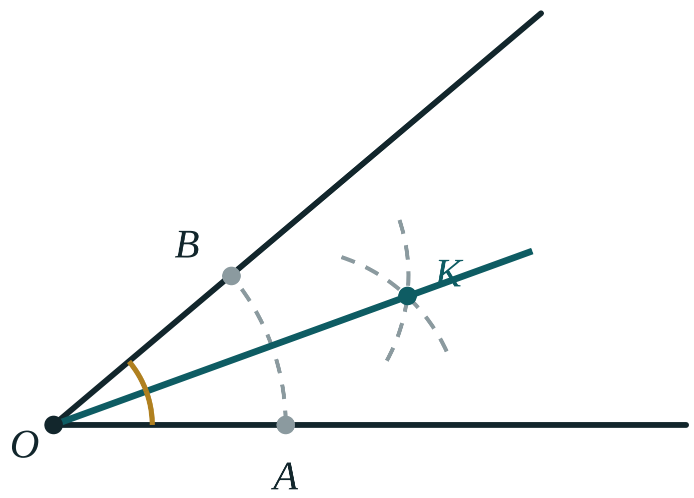
The construction

Through a point A on the line a:

1. with centre A, cut the line at P and Q with one arc;
2. with a larger radius, draw arcs from P and Q meeting at R;
3. $AR \perp a$.

Through a point B not on the line:

1. with centre B, draw an arc cutting the line at P and Q;
2. from P and Q, equal arcs meeting at S on the far side;
3. $BS \perp a$.



Equal arcs from two points give the perpendicular — the same idea in both cases.

> BOTH CONSTRUCTIONS ARE THE PERPENDICULAR BISECTOR
>
> In each case P and Q are equidistant from the new point, so the line through it is the perpendicular bisector of PQ. One construction, learnt once.

The shortest distance

From a point B off a line, the perpendicular BH is shorter than every **oblique** BC to the line.

$BH < BC$ for every $C \neq H$ on the line

Why. In the right triangle BHC, BC is the hypotenuse and BH a leg — and the hypotenuse is always the longest side.

TERM	MEANING
the foot H	where the perpendicular meets the line
the distance from B to a	BH
the projection of BC	HC

“DISTANCE TO A LINE” ALWAYS MEANS THE PERPENDICULAR DISTANCE

Not the distance to a convenient point on it. This convention is used in every later chapter — in the height of a triangle, in the distance to a plane, in the radius of an inscribed circle.

02 Worked examples

EXAMPLE 1 **Two lines cross and one angle is $40°$. Find the other three.**

1. Its adjacent angle: $180° - 40° = 140°$.
2. Its vertical angle: $40°$.
   Vertical angles are equal.
3. The fourth: $140°$.
4. Check: $40 + 140 + 40 + 140 = 360$ ✓

ANSWER

$140°, 40°, 140°$

EXAMPLE 2 **Two adjacent angles are in the ratio $2 : 7$. Find both.**

1. They add to $180°$.
2. Total parts 9.
3. One part = $20°$.
4. $40°$ and $140°$.

ANSWER

$40°$ and $140°$

EXAMPLE 3 **A point B is $5\ cm$ from a line, and C on the line is $12\ cm$ from the foot. Find BC.**

1. $\triangle BHC$ is right-angled at H.
2. $BC^2 = 5^2 + 12^2$
   Pythagoras.
3. $= 169$
4. $BC = 13\ cm$ — longer than $5\ cm$ ✓

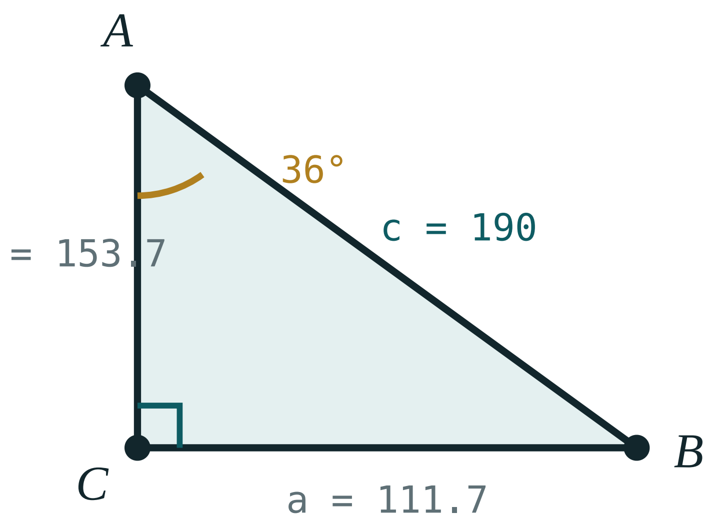
ANSWER

$13\ cm$

03 Interactive model

Ask a pupil to walk from a desk to the wall by the shortest route; they walk perpendicular without being told, and the theorem is confirmed before it is proved.

Perpendicular and oblique



α **36°**	β **54°**	a (opposite) **111.7**	b (adjacent) **153.7**	c (hyp) **190**	sin α **0.59**
cos α **0.81**	tan α **0.73**	$a^2 + b^2$ **36100**	c^2 **36100**		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Perpendicular	Perpendikulyar	Перпендикуляр
Right angle	To‘g‘ri burchak	Прямой угол
Foot of a perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Distance to a line	Chiziqqa masofa	Расстояние до прямой
Adjacent angles	Qo‘shni burchaklar	Смежные углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Oblique	Og‘ma	Наклонная
Set square	Gonio	Угольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a \perp b$ means they meet at:

- $45°$
- $90°$
- $180°$
- any angle

Vertical angles are:

- supplementary
- equal
- complementary
- unrelated

Adjacent angles add to:

- $90°$
- $180°$
- $270°$
- $360°$

One angle at a crossing is $115°$. The vertical one is:

- $65°$
- $115°$
- $245°$
- $90°$

The distance from a point to a line is:

- any segment to it
- the perpendicular
- the longest segment
- undefined

Through a point there is:

- no perpendicular
- exactly one
- two
- many

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. One angle $40°$: its adjacent

   ANSWER $140°$

2. One angle $40°$: its vertical

   ANSWER $40°$

3. One angle $115°$: its adjacent

   ANSWER $65°$

4. One angle $90°$: all four

   ANSWER $90°$

5. Adjacent angles add to

   ANSWER $180°$

6. Vertical angles are

   ANSWER Equal

7. Symbol for perpendicular

   ANSWER $\perp$

Medium · the standard question

7 PROBLEMS

1. Adjacent angles in the ratio $2 : 7$

   ANSWER $40°$ and $140°$

2. Adjacent angles in the ratio $1 : 5$

   ANSWER $30°$ and $150°$

3. One adjacent angle exceeds the other by $40°$

   ANSWER $70°$ and $110°$

4. B is 5 from a line, $HC = 12$: BC

   ANSWER 13

5. B is 9 from a line, $HC = 12$: BC

   ANSWER 15

6. Two crossing lines make four angles totalling

   ANSWER $360°$

7. One angle is three times its adjacent

   ANSWER $45°$ and $135°$

Hard · stretch and reason

7 PROBLEMS

1. The bisectors of two adjacent angles meet at

   ANSWER $90°$

2. Three lines through one point make how many pairs of vertical angles?

   ANSWER 3

3. $BC = 20$ and $BH = 16$: HC

   ANSWER 12

4. A point 8 from a line: the least possible BC

   ANSWER 8

5. Two adjacent angles with bisectors $90°$ apart: always true?

   ANSWER Yes

6. Four angles at a point in the ratio $1 : 2 : 3 : 4$

   ANSWER $36°, 72°, 108°, 144°$

7. Why is the hypotenuse the longest side?

   ANSWER It faces the largest angle

07 Homework

Homework — 5 tasks

Mark every right angle with a small square, and leave construction arcs visible.

1. Two lines cross and one angle is $63°$. Find the other three.
2. Two adjacent angles are in the ratio $3 : 7$. Find both.
3. Construct the perpendicular to a line through a point not on it.
4. A point is $6\ cm$ from a line; a point on the line is $8\ cm$ from the foot. Find the distance between them.

5. Explain why the distance from a point to a line is measured along the perpendicular.

Practical exercises — ordering and multiplying decimals

A Cambridge insert: putting decimals in order, and multiplying them without a calculator.

LESSON 11–12 | 2 × 40 MIN | MATEMATIKA 7, AMALIY MASHQLAR | S8 4.1–4.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Ordering	13 min
Multiplying	23 min
Dividing	22 min
Checking	18 min
Homework	4 min

LEARNING OBJECTIVES

- Order decimals by comparing place value, digit by digit.
- Multiply two decimals and place the point by counting decimal places.
- Divide a decimal by a decimal by making the divisor whole.
- Check every answer by estimation.

TEXTBOOK

National	pp. 32–35
Cambridge	Stage 8, pp. 36–44
Workbook	Exercise 4.1–4.2

01 Explanation

Ordering

Compare decimals digit by digit from the left, not by counting digits.

PAIR	LARGER	WHY
0.7 and 0.65	*0.7*	$7 > 6$ in the tenths
3.09 and 3.1	*3.1*	$1 > 0$ in the tenths
0.4 and 0.400	equal	trailing zeros change nothing

MORE DIGITS DOES NOT MEAN LARGER

0.65 has more digits than *0.7* and is smaller. Writing both with the same number of decimal places — *0.70* and *0.65* — removes the illusion entirely.

Multiplying

STEP	WHAT TO DO
1	ignore the points and multiply as whole numbers
2	count the decimal places in both factors
3	put that many decimal places in the answer

Example. 0.34×0.6. Whole numbers: $34 \times 6 = 204$. Places: $2 + 1 = 3$. Answer: 0.204.

PRODUCT	WHOLE NUMBERS	PLACES	ANSWER
0.3×0.4	12	2	0.12
1.2×0.05	60	3	0.060
2.5×0.4	100	2	1.00

MULTIPLYING TWO DECIMALS BELOW 1 GIVES A SMALLER ANSWER

$0.3 \times 0.4 = 0.12$ — smaller than both. It is the same surprise as division by 0.1, seen from the other side, and the estimate catches it every time.

Dividing

Make the **divisor** a whole number by multiplying both numbers by the same power of ten. The quotient is unchanged.

$$\frac{4.8}{0.06} = \frac{480}{6} = 80$$

DIVISION	REWRITTEN	ANSWER
$7.2 \div 0.9$	$72 \div 9$	8
$0.35 \div 0.7$	$3.5 \div 7$	0.5
$1.44 \div 0.12$	$144 \div 12$	12

BOTH NUMBERS MUST BE MULTIPLIED

Multiplying only the divisor changes the answer. Writing the division as a fraction makes it obvious: a fraction is unchanged only when top and bottom are treated alike.

Checking

CHECK	QUESTION TO ASK
size	is the answer roughly what the estimate said?
direction	should multiplying have made it bigger or smaller?
last digit	does 0.34×0.6 end in the last digit of 4×6?
reverse	does the answer times the divisor give the dividend back?

THE ESTIMATE IS THE CHECK

Every answer in this lesson can be verified in ten seconds by rounding to one significant figure. Doing so consistently is worth more than any amount of careful digit-copying.

02 Worked examples

EXAMPLE 1 **Put in ascending order: $0.7, 0.65, 0.702, 0.6$.**

1. Write with three decimal places: $0.700, 0.650, 0.702, 0.600$.
2. Compare as whole numbers: $600, 650, 700, 702$.
3. $0.6 < 0.65 < 0.7 < 0.702$
4. Equalising the places is the whole method.

ANSWER

$0.6, 0.65, 0.7, 0.702$

EXAMPLE 2 **Compute 0.34×0.6.**

1. $34 \times 6 = 204$

   Whole numbers.

2. Decimal places: $2 + 1 = 3$.

3. 0.204

4. Estimate: $0.3 \times 0.6 = 0.18$ ✓

ANSWER

0.204

EXAMPLE 3 **Compute $4.8 \div 0.06$.**

1. Multiply both by 100.

2. $480 \div 6$

3. $= 80$

4. Check: $80 \times 0.06 = 4.8$ ✓

ANSWER

80

03 Interactive model

Write 0.7 and 0.65 on the board and ask which is larger; collect the wrong answer, then write 0.70 and 0.65 and let the class correct itself.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To order	Tartiblash	Упорядочить
Ascending order	O‘sish tartibi	По возрастанию
Descending order	Kamayish tartibi	По убыванию
Place value	Xona qiymati	Разрядное значение
Divisor	Bo‘luvchi	Делитель
Dividend	Bo‘linuvchi	Делимое
Trailing zero	Oxirgi nol	Конечный ноль
Product	Ko‘paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which is larger, 0.7 or 0.65?

- 0.7
- 0.65
- equal
- cannot tell

0.4 and 0.400 are:

- 0.4 larger
- 0.400 larger
- equal
- different

0.3×0.4 equals:

- 1.2
- 0.12
- 0.012
- 12

0.34×0.6 has how many decimal places?

- 1
- 2
- 3
- 4

$4.8 \div 0.06$ equals:

- 0.8
- 8
- 80
- 800

In a division, you may multiply:

- only the divisor
- only the dividend
- both by the same number
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Larger: 0.7 or 0.65

   ANSWER 0.7

2. Larger: 3.09 or 3.1

   ANSWER 3.1

3. 0.3×0.4

   ANSWER 0.12

4. 0.2×0.5

   ANSWER 0.1

5. $7.2 \div 0.9$

   ANSWER 8

6. $0.35 \div 0.7$

   ANSWER 0.5

7. $1.44 \div 0.12$

   ANSWER 12

Medium · the standard question

7 PROBLEMS

1. Ascending: $0.7, 0.65, 0.702, 0.6$

   ANSWER $0.6, 0.65, 0.7, 0.702$

2. 0.34×0.6

   ANSWER 0.204

3. 1.2×0.05

   ANSWER 0.06

4. 2.5×0.4

   ANSWER 1

5. $4.8 \div 0.06$

   ANSWER 80

6. $0.048 \div 0.8$

   ANSWER 0.06

7. 1.25×0.08

   ANSWER 0.1

Hard · stretch and reason

7 PROBLEMS

1. $0.25 \times 0.4 \times 0.5$

   ANSWER 0.05

2. $(0.2)^3$

   ANSWER 0.008

3. $6.25 \div 0.025$

   ANSWER 250

4. Descending: $0.09, 0.1, 0.089, 0.091$

   ANSWER $0.1, 0.091, 0.09, 0.089$

5. $0.6 \times 0.6 - 0.4 \times 0.9$

   ANSWER 0

6. A cloth of $0.85\ m$ costs $34\ 000$ so‘m: the price per metre

   ANSWER $40\ 000$

7. $12.5 \times 0.008 \div 0.05$

   ANSWER 2

07 Homework

Homework — 5 tasks

Estimate every product and quotient before computing it.

1. Put in ascending order: $0.45, 0.5, 0.409, 0.451$.
2. Compute 0.28×0.7 and 1.5×0.06.
3. Compute $9.6 \div 0.08$ and $0.56 \div 0.7$.
4. Compute $(0.3)^2$ and $(0.1)^3$.
5. $1.2\ kg$ of rice costs $18\ 000$ so‘m. Find the price of $1\ kg$.

Control work 1, and work on the mistakes

The basic geometric figures, tested — and the first chapter closed.

LESSON 13–14 | 2 × 40 MIN | MATEMATIKA 7, NAZORAT ISHI 1 | S8 5 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Use the additive properties of segments and angles under time.
- Classify angles and find complements and supplements.
- Use vertical and adjacent angles at a crossing.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 3–35
Cambridge	Stage 8, pp. 48–56
Workbook	Control paper M1

01 Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	C lies between A and B; $AC = 14$, $AB = 37$: find CB	*4*	L3–4
2	$AB = 26\ cm$: find AM for the midpoint M	*3*	L3–4
3	Classify $35°$, $90°$, $142°$, $200°$	*4*	L5–6
4	Find the complement and supplement of $48°$	*4*	L5–6
5	Two lines cross; one angle is $71°$: find the other three	*5*	L9–10
6	Compute 0.42×0.5 and $3.6 \div 0.04$	*5*	L7–8, 11–12

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for writing $AB = AC + CB$ before substituting; Q3 one for the reflex angle; Q5 two for saying which pairs are vertical; Q6 two for the decimal places.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
subtraction reversed	$CB = 37 + 14$	$CB = 37 - 14$
unit missing	13	$13\ cm$
$200°$	“obtuse”	reflex
complement and supplement swapped	$132°$	$42°$ and $132°$, named correctly
vertical taken as supplementary	$109°$	$71°$
decimal places miscounted	$0.42 \times 0.5 = 2.1$	0.21
only the divisor multiplied	$3.6 \div 4 = 0.9$	$360 \div 4 = 90$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter I as one map

BLOCK	THE SENTENCE
the basic figures	point, line and plane are undefined; everything else follows
the segment	$AB = AC + CB$ when C is between
the midpoint	$AM = MB = \frac{\mathrm{AB}}{2}$
the angle	two rays from one vertex, measured in degrees
the five kinds	acute, right, obtuse, straight, reflex
at a crossing	adjacent add to $180°$, vertical are equal
perpendicular	the shortest distance from a point to a line
decimals	count the places when multiplying; make the divisor whole when dividing

LOOKING FORWARD

The next chapter leaves geometry for a while and introduces **negative numbers** — the first real widening of the number system since fractions. The four operations all have to be learnt again, with signs.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: $AC = 14, AB = 37$.**

1. $AB = AC + CB$

   The formula first.

2. $37 = 14 + CB$

3. $CB = 23$

4. With the unit: $23\ cm$.

ANSWER

$CB = 23\ cm$

EXAMPLE 2 **Model answer, Q5: one angle at a crossing is $71°$.**

1. Adjacent: $180° - 71° = 109°$.
2. Vertical to the first: $71°$.
   Equal.
3. The fourth: $109°$.
4. Total $360°$ ✓

ANSWER

$109°, 71°, 109°$

EXAMPLE 3 **Model answer, Q6: 0.42×0.5 and $3.6 \div 0.04$.**

1. $42 \times 5 = 210$; places $2 + 1 = 3$.
2. $= 0.210 = 0.21$
3. $3.6 \div 0.04 = 360 \div 4$
   Both × 100.
4. $= 90$

ANSWER

0.21 and 90

03 Interactive model

Return the papers with Q3 unmarked and let the class decide together whether 200° is obtuse or reflex; the vocabulary sticks when it is argued over.

Chapter I in twelve questions

Which is undefined?

- segment
- point
- angle
- triangle

Through two points there passes:

- no line
- one line
- two lines
- many

Which has two ends?

- a line
- a ray
- a segment
- a plane

$AB = AC + CB$ needs C:

- anywhere
- between A and B
- outside
- on a ray

A right angle is:

- $45°$
- $90°$
- $180°$
- $360°$

$200°$ is:

- obtuse
- straight
- reflex
- acute

Complementary angles add to:

- $90°$
- $180°$
- $270°$
- $360°$

Vertical angles are:

- supplementary
- equal
- complementary
- unrelated

The distance to a line is measured:

- any way
- along the perpendicular
- to the nearest point drawn
- in degrees

0.3×0.4 equals:

- 1.2
- 0.12
- 0.012
- 12

$6 \div 0.1$ equals:

- 0.6
- 6
- 60
- 600

0.7 and 0.65:

- 0.65 larger
- 0.7 larger
- equal
- cannot tell

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Segment	Kesma	Отрезок
Angle	Burchak	Угол
Midpoint	O‘rta nuqta	Середина
Bisector	Bissektrisa	Биссектриса
Adjacent angles	Qo‘shni burchaklar	Смежные углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 begins with:

- the answer
- the formula
- a guess
- a measurement

Q3 includes which unusual kind?

- acute
- right
- reflex
- straight

In Q5 the vertical angle is:

- $71°$
- $109°$
- $90°$
- $289°$

Q6 needs, for the product:

- an estimate only
- the count of decimal places
- a calculator
- rounding

For the division in Q6, multiply:

- only the divisor
- only the dividend
- both by *100*
- neither

Work on the mistakes means:

- fix the wrong line
- rewrite the solution
- copy the answer
- skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $AC = 14, AB = 37$: CB

ANSWER 23

2. $AB = 26$: AM

ANSWER 13

3. Classify $35°$

ANSWER Acute

4. Classify $142°$

ANSWER Obtuse

5. Classify $200°$

ANSWER Reflex

6. Complement of $48°$

ANSWER $42°$

7. Supplement of $48°$

ANSWER $132°$

Medium · the standard question

7 PROBLEMS

1. One angle $71°$: the other three

ANSWER $109°, 71°, 109°$

2. 0.42×0.5

ANSWER 0.21

3. $3.6 \div 0.04$

ANSWER 90

4. $AB = 40$ with $AC : CB = 3 : 5$: AC

ANSWER 15

5. Adjacent angles in the ratio $1 : 3$

ANSWER $45°$ and $135°$

6. Bisector of $126°$

ANSWER $63°$ each

7. Ascending: $0.5, 0.45, 0.505$

ANSWER $0.45, 0.5, 0.505$

Hard · stretch and reason

7 PROBLEMS

1. An angle is 5 times its complement

   ANSWER $75°$

2. $AB = 36$, M the midpoint of AB, N of MB: AN

   ANSWER 27

3. Four angles at a point in the ratio $2 : 3 : 4 : 3$

   ANSWER $60°, 90°, 120°, 90°$

4. 0.125×0.8

   ANSWER 0.1

5. $2.4 \div 0.015$

   ANSWER 160

6. A point 7 from a line and 24 along it from the foot

   ANSWER 25

7. Two adjacent angles whose bisectors are $90°$ apart: always?

   ANSWER Yes

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the next chapter begins.

1. C lies between A and B with $AC = 19$ and $AB = 45$. Find CB.
2. Find the complement and the supplement of $27°$.
3. Two lines cross and one angle is $124°$. Find the other three.
4. Compute 0.36×0.5 and $4.5 \div 0.09$.
5. Classify $89°$, $90°$, $91°$ and $181°$.

The idea of positive and negative numbers

Numbers below zero — invented so that every subtraction has an answer.

LESSON 15–16 | 2 × 40 MIN | MATEMATIKA 7, §5 | S8 1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Why they are needed	13 min
Reading and writing	23 min
Direction as well as size	22 min
In practice	18 min
Homework	4 min

LEARNING OBJECTIVES

- Explain why negative numbers are needed.
- Read and write negative numbers, and use them in context.
- Interpret a negative number as a direction as well as a size.
- Solve simple problems involving temperature, debt and height.

TEXTBOOK

National	pp. 36–41
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

Why they are needed

Among the numbers known so far, $3 - 5$ has no answer. Every other operation always works; only subtraction can fail.

OPERATION	ALWAYS POSSIBLE IN ℕ?
addition	yes
multiplication	yes
subtraction	**no** — $3 - 5$
division	no — $3 \div 5$ (that gave fractions)

Fractions were invented so that division always works; **negative numbers** are invented so that subtraction always works. The pattern is the same.

THE NUMBER SYSTEM GROWS TO FILL A GAP

Naturals, then fractions, then negatives, then irrationals in Grade 8, then complex numbers in Grade 11 — each new kind of number exists because some ordinary operation had no answer without it.

Reading and writing

WRITTEN	READ	MEANS
-5	minus five	five below zero
$+5$	plus five	five above zero
0	zero	neither positive nor negative
-0.4	minus nought point four	a negative fraction

The $+$ sign is usually left off: 5 and $+5$ are the same number. The $-$ sign is never left off.

ZERO IS NEITHER POSITIVE NOR NEGATIVE

It is the boundary between them, and it belongs to neither side. Statements such as “all numbers are positive or negative” are false because of zero.

Direction as well as size

A negative number carries two pieces of information: a **size** and a **direction**.

SITUATION	+	−
temperature	above zero	below zero
money	owned	owed
height	above sea level	below it
time	after an event	before it
movement	forwards	backwards

WHICH DIRECTION IS POSITIVE IS A CHOICE

Nothing makes “above sea level” positive; it is a convention. What matters is that once chosen it is kept — and that the two directions are opposite.

In practice

STATEMENT	AS A NUMBER
12 degrees of frost	$-12°C$
a debt of $50\ 000$ so‘m	$-50\ 000$
the Dead Sea shore, $430\ m$ below sea level	$-430\ m$
a lift going down 3 floors	-3
a loss of 7 points	-7

Example. The temperature is $-8°C$ and rises by 5 degrees. It becomes $-3°C$ — still below zero. It would need to rise by more than 8 degrees to reach the positive side.

-12 IS COLDER THAN -5

With negatives the bigger-looking number is the smaller one. Reading them on a thermometer, where -12 is plainly lower down, prevents the mistake.

02 Worked examples

EXAMPLE 1 **Write as numbers: 7 degrees of frost; a debt of $30\ 000$; $250\ m$ below sea level.**

1. Frost means below zero: $-7°C$.
2. A debt is money owed: $-30\ 000$.
3. Below sea level: $-250\ m$.
4. Each carries a direction as well as a size.

ANSWER

$-7, -30\ 000, -250$

EXAMPLE 2 **The temperature is $-8°C$ and rises by 5 degrees. Then it falls by 4. Find the final temperature.**

1. A rise of 5: $-8 + 5 = -3$.
   Still below zero.
2. A fall of 4: $-3 - 4$.
3. $= -7°C$
4. Colder than it began.

ANSWER

$-7°C$

EXAMPLE 3 **A lift starts on floor 5, goes down 8 floors, then up 2. Where does it stop?**

1. $5 - 8 = -3$
   Three floors below the ground.
2. $-3 + 2$
3. $= -1$
4. The first basement.

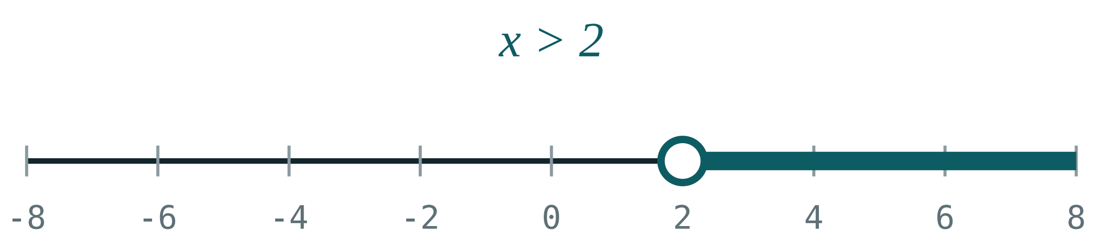
ANSWER

-1 — the first basement

03 Interactive model

Draw a thermometer up the side of the board and mark the day's temperatures; negative numbers arrive as readings, not as an abstraction.

Above and below zero



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Positive number	Musbat son	Положительное число
Negative number	Manfiy son	Отрицательное число
Zero	Nol	Ноль
Sign	Ishora	Знак
Below zero	Noldan past	Ниже нуля
Debt	Qarz	Долг
Sea level	Dengiz sathi	Уровень моря
Rise and fall	Ko‘tarilish va pasayish	Подъём и спад

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Negative numbers were invented so that:

- addition works
- subtraction always works
- division works
- nothing

Zero is:

- positive
- negative
- neither
- both

7 degrees of frost is:

- $7°C$
- $-7°C$
- $70°C$
- $0°C$

A debt of $30\ 000$ is:

- $30\ 000$
- $-30\ 000$
- 0
- not a number

$-8 + 5$ equals:

- -13
- -3
- 3
- 13

Which is colder?

- $-5°C$
- $-12°C$
- equal
- cannot tell

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 9 degrees of frost

   ANSWER -9

2. A debt of $20\ 000$

   ANSWER $-20\ 000$

3. $150\ m$ below sea level

   ANSWER -150

4. A gain of 6 points

   ANSWER $+6$

5. Is 0 positive?

   ANSWER No

6. Is 0 negative?

   ANSWER No

7. Colder: -3 or -9

   ANSWER -9

Medium · the standard question

7 PROBLEMS

1. -8 rises by 5

ANSWER -3

2. -3 falls by 4

ANSWER -7

3. Lift: floor 5, down 8, up 2

ANSWER -1

4. Temperature -15 rises by 20

ANSWER $+5$

5. A balance of $40\ 000$ after spending $55\ 000$

ANSWER $-15\ 000$

6. Higher: $-200\ m$ or $-50\ m$

ANSWER $-50\ m$

7. A submarine at $-80\ m$ rises $35\ m$

ANSWER $-45\ m$

Hard · stretch and reason

7 PROBLEMS

1. The temperature falls from $4°C$ to $-11°C$: the fall

   ANSWER 15 degrees

2. A mountain $3200\ m$ above a lake $-150\ m$: the difference

   ANSWER $3350\ m$

3. Five readings $-3, 0, 2, -5, 1$: the coldest and the warmest

   ANSWER -5 and 2

4. Their range

   ANSWER 7

5. A lift from -2 to 7: floors travelled

   ANSWER 9

6. A debt of $80\ 000$ reduced by $95\ 000$

   ANSWER $+15\ 000$

7. Why is zero neither positive nor negative?

   ANSWER It is the boundary between them

07 Homework

Homework — 5 tasks

Give every answer with its sign and, where there is one, its unit.

1. Write as numbers: 14 degrees of frost; a debt of $45\ 000$; $60\ m$ below sea level.
2. The temperature is $-6°C$ and rises by 9 degrees. Find it now.
3. A lift starts on floor 3, goes down 7, then up 5. Where does it stop?
4. Which is higher: $-120\ m$ or $-45\ m$?
5. The temperature falls from $2°C$ to $-13°C$. By how many degrees?

Home / Grade 7 / Geometry / Quarter 1

The idea of the integers

The naturals, their opposites and zero — one set closed under subtraction.

LESSON 17–18 | 2 × 40 MIN | MATEMATIKA 7, §6 | S8 1.1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The three sets	13 min
Closure	23 min
Factors, multiples, primes	22 min
Recognising integers	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define the set of integers and use the symbol $\mathbb{Z}$.
- Distinguish natural numbers, whole numbers and integers.
- Recall factors, multiples and primes among the naturals.
- Decide whether a given number is an integer.

TEXTBOOK

National	pp. 42–46
Cambridge	Stage 8, pp. 2–6
Workbook	Exercise 1.1

01 Explanation

The three sets

SET	SYMBOL	MEMBERS
natural numbers	$\mathbb{N}$	$1, 2, 3, \dots$
whole numbers	—	$0, 1, 2, 3, \dots$
integers	$\mathbb{Z}$	$\dots, -2, -1, 0, 1, 2, \dots$

Every natural number is an integer, and every integer is a rational number (the next chapter). Written with the subset symbol: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$.

$\mathbb{Z}$ COMES FROM THE GERMAN *ZAHLEN*

Meaning “numbers”. $\mathbb{N}$ is for *naturalis* and $\mathbb{Q}$ for *quoziente*. The letters are international, and they will be used for the rest of school mathematics.

Closure

A set is **closed** under an operation if the answer always stays inside the set.

OPERATION	$\mathbb{N}$	$\mathbb{Z}$
addition	closed	closed
multiplication	closed	closed
subtraction	**not** closed	closed
division	not closed	not closed

The third row is the whole reason for the chapter: $3 - 5$ leaves $\mathbb{N}$ but stays inside $\mathbb{Z}$.
The fourth row is why $\mathbb{Q}$ will be needed.

$6 \div 3 = 2$ DOES NOT MAKE $\mathbb{Z}$ CLOSED UNDER DIVISION

Closure requires **every** case to work. One counter-example — $3 \div 2$ — settles it, just as one counter-example disproved a claim in the algebra course.

Factors, multiples, primes

WORD	MEANS	EXAMPLE
factor of n	divides n exactly	$1, 2, 3, 4, 6, 12$
multiple of n	n times a whole number	$12, 24, 36, \ldots$
prime	exactly two factors	$2, 3, 5, 7, 11, 13$
composite	more than two factors	$4, 6, 8, 9, 10$

1 is neither prime nor composite: it has only one factor. 2 is the only even prime.

EVERY WHOLE NUMBER GREATER THAN 1 IS A PRODUCT OF PRIMES, IN ONE WAY ONLY

$60 = 2^2 \times 3 \times 5$, and no other set of primes gives 60. This is the fundamental theorem of arithmetic, and it underlies HCF, LCM and every fraction calculation.

Recognising integers

NUMBER	AN INTEGER?	WHY
-7	yes	a negative whole number
0	yes	—
$\frac{6}{3}$	yes	it equals 2
$\frac{7}{2}$	no	3.5
-4.0	yes	it equals -4
$0.333\ldots$	no	$\frac{1}{3}$

LOOK AT THE VALUE, NOT AT THE WAY IT IS WRITTEN

$\frac{6}{3}$ is written as a fraction but **is** the integer 2. A number is an integer if it equals one, however it happens to be written.

02 Worked examples

EXAMPLE 1 **Which of $-7, 0, \frac{6}{3}, \frac{7}{2}, -4.0, 0.333\ldots$ are integers?**

1. -7 and 0 plainly are.
2. $\frac{6}{3} = 2$ — an integer.
   Look at the value.
3. $\frac{7}{2} = 3.5$ — not.
4. $-4.0 = -4$ — yes; $0.333\ldots = \frac{1}{3}$ — no.

ANSWER

$-7, 0, \frac{6}{3}, -4.0$

EXAMPLE 2 **List all the factors of 36, and say whether it is prime.**

1. Pair them: $1 \times 36, 2 \times 18, 3 \times 12, 4 \times 9, 6 \times 6$.
2. $1, 2, 3, 4, 6, 9, 12, 18, 36$
   Nine factors.
3. More than two factors.
4. Composite.

ANSWER

Nine factors; composite

EXAMPLE 3 **Is $\mathbb{Z}$ closed under subtraction? Under division?**

1. Subtraction: $a - b$ is always an integer.

   Closed.

2. Division: $3 \div 2 = 1.5$.

3. Not an integer.

4. Closed under subtraction, not under division.

ANSWER

Yes; no

03 Interactive model

Ask the class to find a subtraction of naturals with no natural answer, then to find one of integers with no integer answer; the second search fails, and closure is understood.

Which set does it belong to?

7 is:

- a natural number and an integer
- an integer only
- neither
- negative

-7 is:

- a natural number
- an integer but not natural
- neither
- a fraction

0 is:

- an integer
- not a number
- negative
- natural in every convention

-3.5 is:

- an integer
- not an integer
- natural
- positive

$+5$ and 5 are:

- different numbers
- the same number
- opposites
- both negative

The opposite of -12 is:

- 12
- -12
- 0
- $\frac{1}{12}$

How many integers lie between -2 and 2?

- 2
- 3
- 4
- infinitely many

A temperature of $-8°$ is written with a minus because:

- it is small
- it is below zero
- it is an error
- it is a fraction

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Integer	Butun son	Целое число
Natural number	Natural son	Натуральное число
Set	To‘plam	Множество
Factor	Bo‘luvchi	Делитель
Multiple	Karrali	Кратное
Prime number	Tub son	Простое число
Composite number	Murakkab son	Составное число
Closed under	Yopiq	Замкнутое относительно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\mathbb{Z}$ denotes:

- the naturals
- the integers
- the rationals
- the reals

Is 0 a natural number in this course?

- yes
- no
- sometimes
- undefined

$\mathbb{Z}$ is closed under:

- division
- subtraction
- both
- neither

Is $\frac{6}{3}$ an integer?

- yes
- no
- sometimes
- undefined

How many factors has a prime?

- 1
- 2
- 3
- many

Is 1 prime?

- yes
- no
- sometimes
- undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Is -7 an integer?

   ANSWER Yes

2. Is 3.5 an integer?

   ANSWER No

3. Is $\frac{8}{4}$ an integer?

   ANSWER Yes

4. Is 0 an integer?

   ANSWER Yes

5. Factors of 12

   ANSWER $1, 2, 3, 4, 6, 12$

6. The first five primes

   ANSWER $2, 3, 5, 7, 11$

7. Is 1 prime?

   ANSWER No

Medium · the standard question

7 PROBLEMS

1. Factors of 36

ANSWER $1,2,3,4,6,9,12,18,36$

2. Is 36 prime or composite?

ANSWER Composite

3. Prime factorisation of 60

ANSWER $2^2 \times 3 \times 5$

4. Prime factorisation of 84

ANSWER $2^2 \times 3 \times 7$

5. Is $\mathbb{N}$ closed under subtraction?

ANSWER No

6. Is $\mathbb{Z}$ closed under subtraction?

ANSWER Yes

7. Is $\mathbb{Z}$ closed under division?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. HCF of 36 and 60

ANSWER 12

2. LCM of 36 and 60

ANSWER 180

3. How many factors has 60?

ANSWER 12

4. How many factors has $2^4 \times 3^2$?

ANSWER 15

5. The only even prime

ANSWER 2

6. Every integer is rational: true?

ANSWER Yes — $n = \frac{n}{1}$

7. Is the set of odd numbers closed under addition?

ANSWER No — the sum is even

07 Homework

Homework — 5 tasks

Justify every "no" with a counter-example.

1. Which of $-3, 2.5, \frac{9}{3}, 0, -1.5$ are integers?
2. List all the factors of 48 and say whether it is prime.
3. Write the prime factorisation of 90.
4. Find the HCF and the LCM of 24 and 36.

5. Is the set of even numbers closed under addition? Under division?

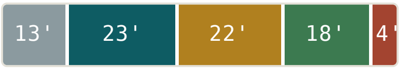
Home / Grade 7 / Geometry / Quarter 1

Positive and negative numbers on the coordinate line

One line, one origin, one unit — and every integer has its own place.

LESSON 19–20 | 2 × 40 MIN | MATEMATIKA 7, §7 | S8 1

LESSON CLOCK



Building the line	13 min
Marking and reading	23 min
Comparing	22 min
Distance	18 min
Homework	4 min

LEARNING OBJECTIVES

- Draw a coordinate line with an origin, a direction and a unit.
- Mark integers and simple fractions on it.
- Read the coordinate of a marked point.
- Use the line to compare numbers and to find distances.

TEXTBOOK

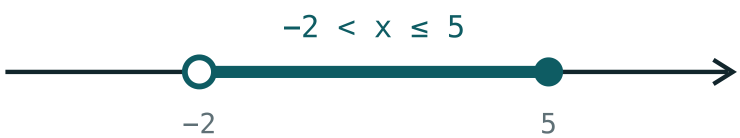
National	pp. 47–52
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

Building the line

A **coordinate line** needs three things:

INGREDIENT	WHAT IT FIXES
an origin O	where 0 is
a positive direction	which way the numbers grow
a unit segment	how far apart consecutive integers are



open circle = strict · filled = included

An origin, a direction and a unit — and every number has a place.

The number attached to a point is its **coordinate**, written $A(-3)$.

EVERY POINT HAS EXACTLY ONE COORDINATE, AND EVERY NUMBER EXACTLY ONE POINT

That one-to-one correspondence is what makes the line useful: a question about numbers becomes a question about positions, and the other way round.

Marking and reading

NUMBER	POSITION
3	3 units right of O
-3	3 units left of O
0	at O
2.5	halfway between 2 and 3
-1.5	halfway between -1 and -2

-1.5 LIES BETWEEN -1 AND -2

Moving left from -1 the numbers become more negative. Reading the negative half as if it were the positive half reversed is the commonest error at this stage.

Comparing

Of two numbers, the one further to the right is the greater

COMPARISON	TRUE STATEMENT	WHY
3 and 5	$3 < 5$	5 is further right
-3 and 5	$-3 < 5$	any negative is less than any positive
-3 and -5	$-5 < -3$	-5 is further left
0 and -2	$-2 < 0$	zero beats every negative

AMONG NEGATIVES, THE ORDER IS REVERSED

$5 > 3$ but $-5 < -3$. The picture makes it obvious; the arithmetic alone does not. Sketching the line before comparing is worth the four seconds it costs.

Distance

The distance between two points is the number of units between them — always positive.

$$AB = |a - b|$$

POINTS	DISTANCE
3 and 7	4
-3 and 5	8
-7 and -2	5
-4 and 0	4

The second row is worth checking on a sketch: from -3 to 0 is 3 units, and from 0 to 5 is 5 more — 8 in all.

SUBTRACT, THEN TAKE THE SIZE

$3 - 7 = -4$, but the distance is 4. A distance is never negative, whichever way the subtraction is done.

02 Worked examples

EXAMPLE 1 **Mark $-4, -1.5, 0, 2, 3.5$ on a coordinate line and list them in ascending order.**

1. Draw the origin, the direction and the unit.
2. -4 is four units left; -1.5 is between -1 and -2.
3. Reading left to right gives the order.
4. $-4 < -1.5 < 0 < 2 < 3.5$

ANSWER

$-4, -1.5, 0, 2, 3.5$

EXAMPLE 2 **Find the distance between the points with coordinates -3 and 5.**

1. $|-3 - 5| = |-8|$
2. $= 8$
3. Check on the line: 3 units to zero, 5 more.
4. 8 units.

ANSWER

8

EXAMPLE 3 **Which is greater, -7 or -2? And -7 or 0.5?**

1. -2 is further right than -7.
2. $-7 < -2$
3. Any negative is less than any positive.
4. $-7 < 0.5$

ANSWER

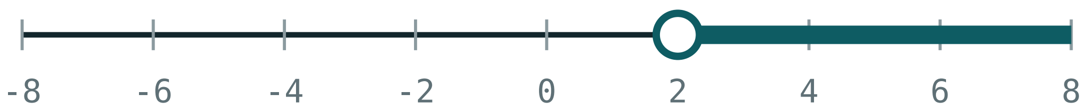
-2 and 0.5

03 Interactive model

Chalk a number line along the floor and have pupils stand on given coordinates; comparing and distance become physical before they are symbolic.

The coordinate line

$$x > 2$$



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate line	Koordinata to‘g‘ri chizig‘i	Координатная прямая
Origin	Boshlanish nuqtasi	Начало отсчёта
Unit segment	Birlik kesma	Единичный отрезок
Coordinate	Koordinata	Координата
Positive direction	Musbat yo‘nalish	Положительное направление
To the right	O‘ngda	Справа
To the left	Chapda	Слева
Distance	Masofa	Расстояние

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A coordinate line needs:

- an origin only
- an origin, a direction and a unit
- a unit only
- two points

-1.5 lies between:

- -1 and 0
- -1 and -2
- 1 and 2
- 0 and 1

The greater of -3 and -5:

- -3
- -5
- equal
- cannot tell

Any negative number is:

- greater than every positive
- less than every positive
- equal to zero
- undefined

The distance between -3 and 5:

- 2
- 8
- -8
- 15

A distance is:

- sometimes negative
- never negative
- always zero
- a coordinate

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Greater: 3 or 5

   ANSWER 5

2. Greater: -3 or 5

   ANSWER 5

3. Greater: -3 or -5

   ANSWER -3

4. Greater: 0 or -2

   ANSWER 0

5. Distance from 3 to 7

   ANSWER 4

6. Distance from -7 to -2

   ANSWER 5

7. Distance from -4 to 0

   ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Ascending: $-4, -1.5, 0, 2, 3.5$

ANSWER As written

2. Ascending: $2, -7, 0, -1, 5$

ANSWER $-7, -1, 0, 2, 5$

3. Distance from -3 to 5

ANSWER 8

4. Distance from -12 to 4

ANSWER 16

5. The point halfway between -6 and 2

ANSWER -2

6. The point halfway between -9 and -1

ANSWER -5

7. Descending: $-2, -8, 3, 0$

ANSWER $3, 0, -2, -8$

Hard · stretch and reason

7 PROBLEMS

1. All integers x with $-4 < x \leq 2$

   ANSWER $-3, -2, -1, 0, 1, 2$

2. How many integers between -15 and 9?

   ANSWER 23

3. Points 5 units from -2

   ANSWER 3 and -7

4. A point twice as far from 0 as -4, on the same side

   ANSWER -8

5. The midpoint of a and b

   ANSWER $\frac{a+b}{2}$

6. Distance between a and $-a$

   ANSWER $2|a|$

7. All x with distance 3 from 7

   ANSWER 4 and 10

07 Homework

Homework — 5 tasks

Sketch the line for every comparison; the picture is the argument.

1. Mark $-5, -2.5, 0, 1, 4$ on a coordinate line.
2. Put in ascending order: $3, -6, 0, -1, 2$.
3. Find the distance between -8 and 3.
4. Find the point halfway between -10 and 4.

5. List all integers x with $-5 \leq x < 3$.

Opposite numbers, and the modulus of a number

The same distance from zero, on the two sides — and the number that forgets the side.

LESSON 21–22 | 2 × 40 MIN | MATEMATIKA 7, §8 | S8 1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Opposite numbers	13 min
The modulus	23 min
Equations with a modulus	22 min
Modulus and distance	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define opposite numbers and use −(−a) = a.
- Define the modulus and compute it for any number.
- Solve simple equations of the form |x| = a.
- Use the modulus to express a distance on the coordinate line.

TEXTBOOK

National	pp. 53–58
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

Opposite numbers

Two numbers are **opposite** if they lie the same distance from 0 on opposite sides. The opposite of a is written $-a$.

NUMBER	ITS OPPOSITE
7	-7
-7	7
0	0
-2.5	2.5

$$-(-a) = a \quad a + (-a) = 0$$

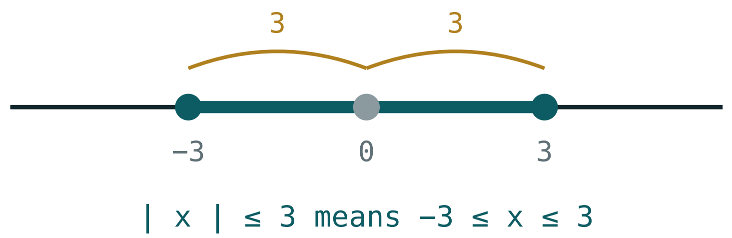
$-A$ IS NOT ALWAYS A NEGATIVE NUMBER

If $a = -7$ then $-a = 7$, which is positive. The minus sign means “the opposite of”, not “a negative number”. This is one of the hardest points of the whole chapter.

The modulus

The **modulus** $|a|$ is the distance from a to 0 on the coordinate line. Distance is never negative, so:

$$|a| = a \text{ if } a \geq 0, \ |a| = -a \text{ if } a < 0$$



The modulus is the distance to the origin — the same for a number and its opposite.

A	$\lvert A\rvert$
7	7
-7	7
0	0
-2.5	2.5

THE MODULUS FORGETS THE SIGN, AND KEEPS THE SIZE

Opposite numbers have the same modulus, which is exactly what “same distance from zero” means. So $|a| = |-a|$ for every a.

Equations with a modulus

EQUATION	SOLUTIONS	WHY
$\|x\| = 5$	$x = 5$ or $x = -5$	two points at distance 5
$\|x\| = 0$	$x = 0$	only the origin
$\|x\| = -3$	none	a distance is never negative
$\|x\| < 3$	$-3 < x < 3$	nearer than 3 to zero

$|X| = 5$ HAS TWO ANSWERS, NOT ONE

Giving only $x = 5$ loses half the marks. Asking “which points are 5 units from zero?” makes both visible immediately.

Modulus and distance

The distance between the points with coordinates a and b is

$$d = |a - b| = |b - a|$$

POINTS	WORKING	DISTANCE
$3, 10$	$\|3 - 10\| = \|-7\|$	7
$-4, 6$	$\|-4 - 6\| = \|-10\|$	10
$-9, -2$	$\|-9 + 2\| = \|-7\|$	7

The order of subtraction does not matter, because $|a - b|$ and $|b - a|$ are opposites and therefore have the same modulus.

TWO USEFUL PROPERTIES

$|ab| = |a| \cdot |b|$ and $|a| \geq 0$ always, with equality only for $a = 0$. Both follow at once from the definition, and both will be used in every later year.

02 Worked examples

EXAMPLE 1 **Find $|-7|$, $|3.5|$, $|0|$ and $-(-4)$.**

1. $|-7| = 7$

   Seven units from zero.

2. $|3.5| = 3.5$

3. $|0| = 0$

4. $-(-4) = 4$

   The opposite of the opposite.

ANSWER

$7, 3.5, 0, 4$

EXAMPLE 2 **Solve $|x| = 6$ and $|x| = -2$.**

1. Which points are 6 from zero?

2. $x = 6$ or $x = -6$.

   Two answers.

3. A distance is never negative.

4. $|x| = -2$ has no solution.

ANSWER

$x = \pm 6$; no solution

EXAMPLE 3 **Find the distance between the points -4 and 6 on the coordinate line.**

1. $d = |-4 - 6|$
2. $= |-10|$
3. $= 10$
4. Check: 4 units to zero, then 6 more.

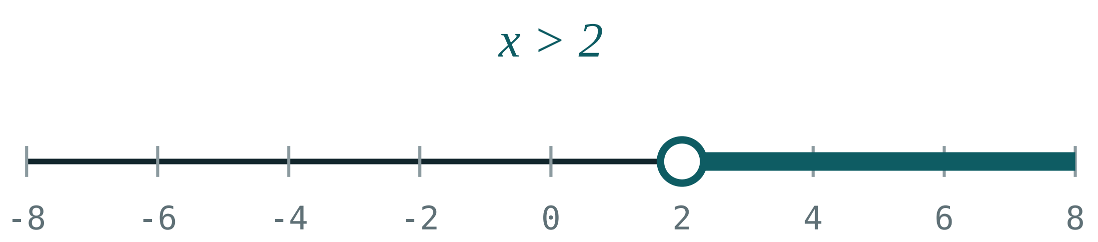
ANSWER

10

03 Interactive model

Ask two pupils to stand equally far from a chalk zero on opposite sides; they have opposite coordinates and the same modulus, and the definition needs no more.

Distance from zero



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Opposite number	Qarama-qarshi son	Противоположное число
Modulus	Modul	Модуль
Absolute value	Absolyut qiymat	Абсолютная величина
Distance from zero	Noldan masofa	Расстояние от нуля
Non-negative	Manfiy emas	Неотрицательный
Two solutions	Ikkita yechim	Два решения
Sign	Ishora	Знак
Symmetric	Simmetrik	Симметричный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The opposite of -7:

- -7
- 7
- 0
- $\frac{1}{7}$

$-(-a)$ equals:

- $-a$
- a
- 0
- $|a|$

$|-7|$ equals:

- -7
- 7
- 0
- 49

$|x| = 5$ has:

- one solution
- two solutions
- no solution
- many

$|x| = -3$ has:

- one solution
- two solutions
- no solution
- $x = 3$

The distance between a and b:

- $a - b$
- $|a - b|$
- $a + b$
- $|a| - |b|$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Opposite of 7

   ANSWER -7

2. Opposite of -7

   ANSWER 7

3. Opposite of 0

   ANSWER 0

4. $|-7|$

   ANSWER 7

5. $|3.5|$

   ANSWER 3.5

6. $|0|$

   ANSWER 0

7. $-(-4)$

   ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Solve $|x| = 6$

ANSWER $x = \pm 6$

2. Solve $|x| = 0$

ANSWER $x = 0$

3. Solve $|x| = -2$

ANSWER No solution

4. Distance between -4 and 6

ANSWER 10

5. Distance between -9 and -2

ANSWER 7

6. $|-3| + |5|$

ANSWER 8

7. $|-3 + 5|$

ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. Solve $|x| < 3$ in integers

ANSWER $-2, -1, 0, 1, 2$

2. Solve $|x| \ge 4$ in integers from -6 to 6

ANSWER $\pm 4, \pm 5, \pm 6$

3. $|-2| \cdot |-5|$

ANSWER 10

4. $|(-2)(-5)|$

ANSWER 10

5. For which a is $|a| = a$?

ANSWER $a \ge 0$

6. For which a is $|a| = -a$?

ANSWER $a \le 0$

7. Solve $|x - 3| = 5$

ANSWER $x = 8$ or $x = -2$

07 Homework

Homework — 5 tasks

Every modulus question has a picture: mark the point and measure to zero.

1. Find $|-12|$, $|4.8|$, $-(-9)$ and the opposite of -15.
2. Solve $|x| = 9$ and $|x| = -1$.
3. Find the distance between -11 and 5.
4. List all integers with $|x| \le 3$.
5. Explain why $-a$ need not be a negative number.

Home / Grade 7 / Geometry / Quarter 1

Comparing integers

Two rules, both read off the coordinate line — and one trap with negatives.

LESSON 23–24 | 2 × 40 MIN | MATEMATIKA 7, §9 | S8 1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The rules	13 min
Two negatives	23 min
Ordering	22 min
The symbols	18 min
Homework	4 min

LEARNING OBJECTIVES

- Compare any two integers using their positions on the line.
- Compare two negatives by comparing their moduli, in reverse.
- Order a list of integers, ascending and descending.
- Use the symbols <, >, ≤, ≥ correctly.

TEXTBOOK

National	pp. 59–63
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

The rules

CASE	RULE	EXAMPLE
both positive	as usual	$3 < 8$
one of each	the negative is smaller	$-3 < 8$
a negative and zero	the negative is smaller	$-3 < 0$
zero and a positive	zero is smaller	$0 < 8$
both negative	the one with the larger modulus is smaller	$-8 < -3$

Every row is the same statement about the line: further right means greater.

ONE PICTURE REPLACES FIVE RULES

Sketch the line, mark both numbers, read left to right. The table above is only a record of what the picture shows.

Two negatives

This is the only case that catches people out.

> For negatives: the bigger the modulus, the smaller the number

PAIR	MODULI	COMPARISON
$-3, -8$	$3, 8$	$-8 < -3$
$-15, -2$	$15, 2$	$-15 < -2$
$-1, -100$	$1, 100$	$-100 < -1$

A debt of 100 is a worse position than a debt of 1; a temperature of $-100°$ is colder than $-1°$. The rule matches every real situation.

$-8 > -3$ IS FALSE

Comparing the moduli and forgetting to reverse is the standard mistake. Saying the numbers aloud as temperatures settles it in a second.

Ordering

Ascending means smallest first; **descending** means largest first.

LIST	ASCENDING
$3, -7, 0, -1, 5$	$-7, -1, 0, 3, 5$
$-2, -9, -5$	$-9, -5, -2$
$10, -10, 1, -1$	$-10, -1, 1, 10$

A method that never fails: put the negatives first, ordered by decreasing modulus, then zero, then the positives ordered as usual.

SAY THE LIST AS TEMPERATURES

“Minus nine, minus five, minus two, zero, three, five” is plainly a list of increasing warmth. The language does the ordering for you.

The symbols

SYMBOL	READ	TRUE EXAMPLE
$<$	is less than	$-5 < -2$
$>$	is greater than	$3 > -7$
$\leq$	is less than or equal to	$4 \leq 4$
$\geq$	is greater than or equal to	$-2 \geq -2$
$\neq$	is not equal to	$3 \neq -3$

A double inequality such as $-3 < x \leq 2$ says two things at once, and is read as a stretch of the line between two marks.

THE POINT OF THE SYMBOL FACES THE SMALLER NUMBER

$-5 < -2$: the narrow end is at -5. Reading the symbol as a mouth that eats the larger number is a reliable check.

02 Worked examples

EXAMPLE 1 **Compare -8 and -3, and -8 and 3.**

1. Both negative: compare moduli 8 and 3.
2. Larger modulus, smaller number: $-8 < -3$.
3. A negative is always less than a positive.
4. $-8 < 3$

ANSWER

$-8 < -3$ and $-8 < 3$

EXAMPLE 2 **Put $3, -7, 0, -1, 5$ in ascending order.**

1. Negatives first, by decreasing modulus: $-7, -1$.
2. Then zero.
3. Then the positives: $3, 5$.
4. $-7, -1, 0, 3, 5$

ANSWER

$-7, -1, 0, 3, 5$

EXAMPLE 3 **List all integers x with $-4 \le x < 2$.**

1. $\le$ includes -4.
2. $<$ excludes 2.
3. Count from -4 up to 1.
4. $-4, -3, -2, -1, 0, 1$

   Six integers.

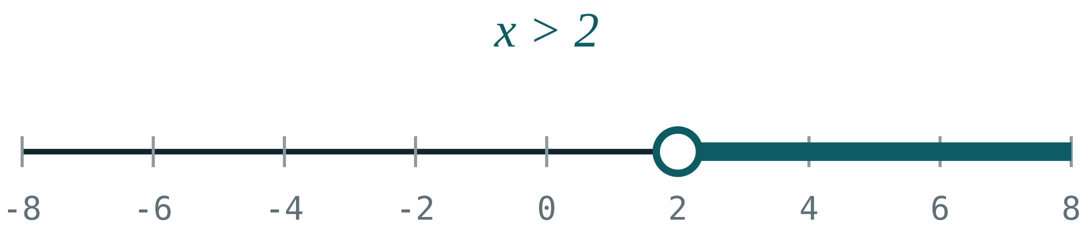
ANSWER

$-4, -3, -2, -1, 0, 1$

03 Interactive model

Read a week of winter temperatures aloud and ask which day was coldest; the ordering of negatives is settled by experience before it is a rule.

Ordering on the line



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To compare	Taqqoslash	Сравнить
Greater than	Katta	Больше
Less than	Kichik	Меньше
Ascending order	O‘sish tartibi	По возрастанию
Descending order	Kamayish tartibi	По убыванию
Inequality sign	Tengsizlik belgisi	Знак неравенства
Between	Orasida	Между
Consecutive	Ketma-ket	Последовательные

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which is greater, -8 or -3?

- -8
- -3
- equal
- cannot tell

A negative number compared with a positive is:

- greater
- less
- equal
- it depends

For negatives, a larger modulus means:

- a larger number
- a smaller number
- the same
- nothing

Ascending order of $-2, -9, -5$:

- $-2, -5, -9$
- $-9, -5, -2$
- $-5, -2, -9$
- $-9, -2, -5$

$-2 \geq -2$ is:

- true
- false
- meaningless
- sometimes

Integers with $-4 \leq x < 2$ number:

- 5
- 6
- 7
- 4

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Greater: -8 or -3

   ANSWER -3

2. Greater: -8 or 3

   ANSWER 3

3. Greater: 0 or -1

   ANSWER 0

4. Greater: -100 or -1

   ANSWER -1

5. Is $-5 < -2$?

   ANSWER True

6. Is $-5 > -2$?

   ANSWER False

7. Is $4 \leq 4$?

   ANSWER True

Medium · the standard question

7 PROBLEMS

1. Ascending: $3, -7, 0, -1, 5$

   ANSWER $-7, -1, 0, 3, 5$

2. Ascending: $-2, -9, -5$

   ANSWER $-9, -5, -2$

3. Descending: $10, -10, 1, -1$

   ANSWER $10, 1, -1, -10$

4. Integers with $-4 \leq x < 2$

   ANSWER $-4, -3, -2, -1, 0, 1$

5. Integers with $-2 < x \leq 3$

   ANSWER $-1, 0, 1, 2, 3$

6. The greatest integer less than -3

   ANSWER -4

7. The least integer greater than -3

   ANSWER -2

Hard · stretch and reason

7 PROBLEMS

1. How many integers satisfy $-20 < x < 15$?

   ANSWER 34

2. The greatest negative integer

   ANSWER -1

3. The least positive integer

   ANSWER 1

4. Order $-|-4|, |-3|, -(-2), 0$

   ANSWER $-4, 0, 2, 3$

5. If $a < 0$ and $b > 0$, compare a and $-b$

   ANSWER Cannot tell without more

6. Between which two consecutive integers does -3.7 lie?

   ANSWER -4 and -3

7. The smallest three-digit negative integer

   ANSWER -999

07 Homework

Homework — 5 tasks

Sketch the line for every comparison, and say each list aloud as temperatures.

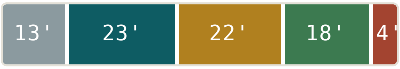
1. Compare -11 and -4, and -11 and 4.
2. Put in ascending order: $-6, 2, 0, -13, 7$.
3. Put in descending order: $-1, -8, 5, 0$.
4. List all integers x with $-5 < x \le 1$.
5. Name the greatest negative integer and the least positive integer.

Addition of integers

Two rules — same signs and different signs — and a number line that explains both.

LESSON 25–26 | 2 × 40 MIN | MATEMATIKA 7, §10 | S8 1.2

LESSON CLOCK



On the line	13 min
Same signs	23 min
Different signs	22 min
Several terms	18 min
Homework	4 min

LEARNING OBJECTIVES

- Add two integers with the same sign and with different signs.
- Use the coordinate line as a model for addition.
- Use the commutative and associative laws to add several integers.
- Solve word problems involving gains and losses.

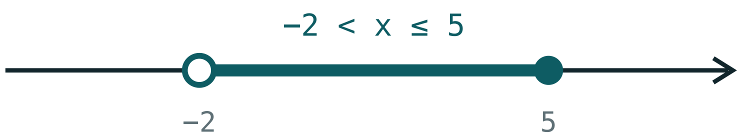
TEXTBOOK

National	pp. 64–70
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

On the line

To add b to a: start at a and move $|b|$ units — right if b is positive, left if it is negative.



Adding a positive number moves right; adding a negative one moves left.

SUM	START	MOVE	END
$3 + 4$	3	4 right	7
$3 + (-4)$	3	4 left	-1
$-3 + 4$	-3	4 right	1
$-3 + (-4)$	-3	4 left	-7

THE LINE IS THE DEFINITION, THE RULES ARE THE SHORTCUT

Anyone who forgets a rule can rebuild it in five seconds on a sketch. That is why the picture comes first in this lesson and the rules second.

Same signs

Add the moduli and keep the common sign

SUM	MODULI	ANSWER
$5 + 8$	$5 + 8 = 13$	13
$(-5) + (-8)$	$5 + 8 = 13$	-13
$(-12) + (-7)$	$12 + 7 = 19$	-19

Two debts make a bigger debt; two falls in temperature make a bigger fall. The rule matches every situation.

Different signs

Subtract the smaller modulus from the larger, and keep the sign of the larger modulus

SUM	MODULI	LARGER	ANSWER
$9 + (-4)$	$9, 4$	9	5
$4 + (-9)$	$4, 9$	9	-5
$(-12) + 7$	$12, 7$	12	-5
$6 + (-6)$	$6, 6$	equal	0

THE SIGN COMES FROM THE LARGER MODULUS, NOT FROM THE FIRST NUMBER

$4 + (-9) = -5$, even though the sum starts with a positive. Deciding the sign first, then the size, keeps the two steps separate.

Several terms

Addition is **commutative** and **associative**, so the terms may be reordered and regrouped freely.

$$a + b = b + a \quad (a + b) + c = a + (b + c)$$

Method. Add all the positives, add all the negatives, then combine the two results.

SUM	POSITIVES	NEGATIVES	ANSWER
$7 - 3 + 5 - 12$	12	-15	-3
$-4 + 9 - 6 + 1$	10	-10	0
$-2 - 5 - 8$	0	-15	-15

TWO PILES, ONE SUBTRACTION

Grouping like signs turns a long chain into a single easy step. It is the same idea as collecting like terms in algebra, which arrives later this year.

02 Worked examples

EXAMPLE 1 **Compute $(-5) + (-8)$, $9 + (-4)$ and $4 + (-9)$.**

1. Same signs: add moduli, keep the sign — -13.
2. Different signs: $9 - 4 = 5$, sign of 9 — 5.
3. Different signs: $9 - 4 = 5$, sign of -9 — -5.
4. Sign first, then size.

ANSWER

$-13, 5, -5$

EXAMPLE 2 **Compute $7 - 3 + 5 - 12$.**

1. Positives: $7 + 5 = 12$.
2. Negatives: $-3 - 12 = -15$.
3. $12 + (-15)$
4. $= -3$

ANSWER

-3

EXAMPLE 3

A shop makes a profit of *45 000*, then a loss of *70 000*, then a profit of *30 000*. Find the net result.

1. Profits: $45\ 000 + 30\ 000 = 75\ 000$.
2. Loss: $-70\ 000$.
3. $75\ 000 + (-70\ 000)$
4. $= +5000$ — a small profit.

ANSWER

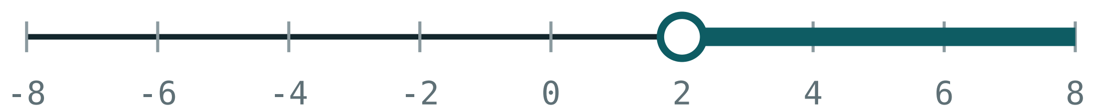
A profit of 5000

03 Interactive model

Do every first example twice — once by the rule and once by walking the line on the floor; the two answers agreeing is what makes the rule trustworthy.

Adding on the line

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Addition	Qo‘shish	Сложение
Sum	Yig‘indi	Сумма
Same sign	Bir xil ishorali	Одинаковые знаки
Different signs	Turli ishorali	Разные знаки
Commutative law	O‘rin almashtirish qonuni	Переместительный закон
Associative law	Guruhlash qonuni	Сочетательный закон
Term	Qo‘shiluvchi	Слагаемое
Gain and loss	Foyda va zarar	Прибыль и убыток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(-5) + (-8)$ equals:

- 13
- -13
- 3
- -3

$9 + (-4)$ equals:

- 13
- -13
- 5
- -5

$4 + (-9)$ equals:

- 5
- -5
- 13
- -13

$6 + (-6)$ equals:

- 12
- -12
- 0
- 6

For different signs you:

- add the moduli
- subtract the moduli
- multiply them
- ignore them

Addition is:

- commutative only
- associative only
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3 + 4$

ANSWER 7

2. $3 + (-4)$

ANSWER -1

3. $-3 + 4$

ANSWER 1

4. $-3 + (-4)$

ANSWER -7

5. $(-5) + (-8)$

ANSWER -13

6. $9 + (-4)$

ANSWER 5

7. $6 + (-6)$

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. $(-12) + 7$

ANSWER -5

2. $4 + (-9)$

ANSWER -5

3. $7 - 3 + 5 - 12$

ANSWER -3

4. $-4 + 9 - 6 + 1$

ANSWER 0

5. $-2 - 5 - 8$

ANSWER -15

6. $15 + (-15) + 8$

ANSWER 8

7. Profit 45, loss 70, profit 30

ANSWER $+5$

Hard · stretch and reason

7 PROBLEMS

1. $-17 + 25 - 8 + 3 - 11$

   ANSWER -8

2. $(-3) + (-3) + (-3) + (-3)$

   ANSWER -12

3. A temperature of $-6°$ rises 4, falls 9, rises 12

   ANSWER $+1°$

4. Find x: $x + (-7) = 3$

   ANSWER 10

5. Find x: $-5 + x = -12$

   ANSWER -7

6. The sum of all integers from -10 to 10

   ANSWER 0

7. The sum of all integers from -10 to 12

   ANSWER 23

07 Homework

Homework — 5 tasks

Decide the sign of the answer before computing its size.

1. Compute $(-7) + (-6)$, $11 + (-4)$ and $5 + (-13)$.
2. Compute $9 - 4 + 7 - 15$.
3. Compute $-3 - 8 + 20 - 6$.
4. A bank balance of $60\ 000$ has $85\ 000$ withdrawn and $40\ 000$ paid in. Find the balance.
5. Find x if $x + (-9) = -2$.

Subtraction of integers

Subtracting is adding the opposite — one rule that replaces the whole operation.

LESSON 27–28 | 2 × 40 MIN | MATEMATIKA 7, §11 | S8 1.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

One rule	13 min
Two signs in a row	23 min
Chains	22 min
Differences	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and use $a - b = a + (-b)$.
- Simplify expressions containing two signs in a row.
- Compute chains of additions and subtractions.
- Use subtraction to find a difference or a change.

TEXTBOOK

National	pp. 71–76
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

One rule

$$a - b = a + (-b)$$

To subtract a number, add its opposite. Subtraction is not a new operation at all — it is addition, once the opposite is taken.

SUBTRACTION	AS AN ADDITION	ANSWER
$7 - 3$	$7 + (-3)$	4
$3 - 7$	$3 + (-7)$	-4
$-3 - 7$	$-3 + (-7)$	-10
$-3 - (-7)$	$-3 + 7$	4
$3 - (-7)$	$3 + 7$	10

NOTHING NEW HAS TO BE LEARNT

Every subtraction becomes an addition, and addition was the last lesson. That is why the rule for subtraction is a single line while the rule for addition took two cases.

Two signs in a row

WRITTEN	MEANS	EXAMPLE
$+(+a)$	$+a$	$5 + (+3) = 8$
$+(-a)$	$-a$	$5 + (-3) = 2$
$-(+a)$	$-a$	$5 - (+3) = 2$
$-(-a)$	$+a$	$5 - (-3) = 8$

Two like signs give a plus; two unlike signs give a minus.

$5 - (-3)$ IS 8, NOT 2

Taking away a debt leaves you better off. The everyday sentence explains the arithmetic, and it is worth saying aloud each time until the rule is automatic.

Chains

Rewrite every subtraction as an addition first, then use the two-pile method.

Example. $-4 - (-9) + (-6) - 2$.

STEP	EXPRESSION
rewrite	$-4 + 9 - 6 - 2$
positives	9
negatives	$-4 - 6 - 2 = -12$
combine	$9 - 12 = -3$

REWRITE THE WHOLE CHAIN BEFORE COMPUTING ANYTHING

Doing the sign work and the arithmetic at the same time is where errors come from. Two separate passes are quicker in the end.

Differences

The **change** from a to b is $b - a$: the new value minus the old one. A positive answer is a rise, a negative one a fall.

FROM	TO	CHANGE	MEANING
$4°$	$11°$	$+7$	a rise of 7
$4°$	$-3°$	-7	a fall of 7
$-9°$	$-2°$	$+7$	a rise of 7
$-2°$	$-9°$	-7	a fall of 7

“NEW MINUS OLD”, IN THAT ORDER

Reversing the subtraction reverses the sign and turns a rise into a fall. Writing the words “new − old” at the top of the working prevents it.

02 Worked examples

EXAMPLE 1 **Compute $3-(-7)$, $-3-7$ and $-3-(-7)$.**

1. $3-(-7) = 3+7 = 10$

   Two minuses give a plus.

2. $-3-7 = -3+(-7) = -10$

3. $-3-(-7) = -3+7 = 4$

4. Rewrite first, compute second.

ANSWER

$10, -10, 4$

EXAMPLE 2 **Compute $-4-(-9)+(-6)-2$.**

1. Rewrite: $-4+9-6-2$.
2. Positives: 9.
3. Negatives: -12.
4. $9-12=-3$

ANSWER

-3

EXAMPLE 3 **The temperature falls from $4°C$ to $-9°C$. Find the change and the size of the fall.**

1. Change $= new - old = -9-4$.
2. $= -13$
3. Negative, so it is a fall.
4. A fall of 13 degrees.

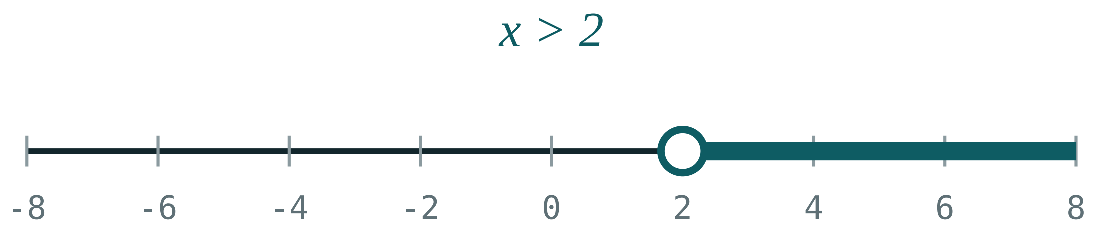
ANSWER

-13 — a fall of $13°$

03 Interactive model

Say “taking away a debt makes you richer” once, and then let the class supply it themselves every time a −(−) appears; the rule sticks as a sentence, not a symbol.

Subtracting on the line



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Subtraction	Ayirish	Вычитание
Difference	Ayirma	Разность
Minuend	Kamayuvchi	Уменьшаемое
Subtrahend	Ayiriluvchi	Вычитаемое
Opposite	Qarama-qarshi	Противоположное
Two signs in a row	Ketma-ket ikki ishora	Два знака подряд
Change	O‘zgarish	Изменение
Rise and fall	Ko‘tarilish va pasayish	Подъём и спад

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a - b$ equals:

- $a + b$
- $a + (-b)$
- $-a + b$
- $b - a$

$5 - (-3)$ equals:

- 2
- 8
- -2
- -8

$-3 - 7$ equals:

- 4
- -4
- 10
- -10

$-(-a)$ equals:

- $-a$
- a
- 0
- $|a|$

The change from 4 to -3:

- 7
- -7
- 1
- -1

A negative change means:

- a rise
- a fall
- no change
- an error

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $7 - 3$

   ANSWER 4

2. $3 - 7$

   ANSWER -4

3. $-3 - 7$

   ANSWER -10

4. $3 - (-7)$

   ANSWER 10

5. $-3 - (-7)$

   ANSWER 4

6. $5 - (+3)$

   ANSWER 2

7. $5 + (-3)$

   ANSWER 2

Medium · the standard question

7 PROBLEMS

1. $-4 - (-9) + (-6) - 2$

ANSWER -3

2. $12 - 20 + 5 - (-3)$

ANSWER 0

3. $-15 - (-15)$

ANSWER 0

4. $0 - (-8)$

ANSWER 8

5. Change from $4°$ to $-9°$

ANSWER -13

6. Change from $-9°$ to $-2°$

ANSWER $+7$

7. $-7 - (-2) - 5$

ANSWER -10

Hard · stretch and reason

7 PROBLEMS

1. $-3 - (-5) - (-7) - 9$

   ANSWER 0

2. Find x: $x - (-4) = 1$

   ANSWER -3

3. Find x: $-6 - x = 2$

   ANSWER -8

4. The difference between the highest and lowest of $-12, 5, -3, 8$

   ANSWER 20

5. A lift from -3 to 9: floors travelled

   ANSWER 12

6. On a day the temperature goes $-7 \to 3 \to -5$

   ANSWER A rise of 10, a fall of 8

7. Simplify $a - (-b) - (-c)$

   ANSWER $a + b + c$

07 Homework

Homework — 5 tasks

Rewrite every subtraction as an addition before computing anything.

1. Compute $8 - (-5)$, $-8 - 5$ and $-8 - (-5)$.
2. Compute $-6 - (-11) + (-4) - 3$.
3. The temperature falls from $7°C$ to $-6°C$. Find the change.
4. Find the difference between the highest and lowest of $-15, 4, -8, 11$.
5. Find x if $x - (-6) = -1$.

Multiplication of integers

Multiply the moduli, then decide the sign — and two minuses really do make a plus.

LESSON 29–30 | 2 × 40 MIN | MATEMATIKA 7, §12 | S8 1.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The sign rules	13 min
Why two minuses make a plus	23 min
Several factors	22 min
Powers	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and use the sign rules for multiplication.
- Explain why (−a)(−b) = ab.
- Find the sign of a product of several factors.
- Compute powers of negative numbers.

TEXTBOOK

National	pp. 77–82
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

The sign rules

> Multiply the moduli; the product is positive if the signs are alike and negative if they differ

SIGNS	PRODUCT	EXAMPLE
$(+)(+)$	$+$	$4 \times 5 = 20$
$(+)(-)$	$-$	$4 \times (-5) = -20$
$(-)(+)$	$-$	$(-4) \times 5 = -20$
$(-)(-)$	$+$	$(-4) \times (-5) = 20$
anything $\times\, 0$	0	$(-4) \times 0 = 0$

SIGN FIRST, THEN SIZE

Decide the sign from the two signs alone, write it down, then multiply the moduli. Two separate decisions are far more reliable than one combined one.

Why two minuses make a plus

Look at a pattern in which one factor decreases by 1 each time:

PRODUCT	VALUE
$(-4) \times 3$	-12
$(-4) \times 2$	-8
$(-4) \times 1$	-4
$(-4) \times 0$	0
$(-4) \times (-1)$	4
$(-4) \times (-2)$	8

Each step adds 4. Continuing the pattern past zero forces the products to become positive — there is no other way to keep the arithmetic consistent.

THE RULE IS NOT A CONVENTION, IT IS FORCED

If $(-4)(-1)$ were -4, then the distributive law would fail. Mathematics chooses the only definition under which all the old laws keep working.

Several factors

> The product is negative when the number of negative factors is odd, and positive when it is even

PRODUCT	NEGATIVE FACTORS	SIGN	VALUE
$(-2)(-3)(-4)$	3	−	-24
$(-2)(-3)(4)$	2	+	24
$(-1)(-1)(-1)(-1)$	4	+	1
$(-5)(2)(-3)(-1)$	3	−	-30

ONE ZERO FACTOR MAKES THE WHOLE PRODUCT ZERO

However many negatives there are. Scanning for a zero before counting signs saves the whole calculation.

Powers

POWER	MEANING	SIGN	VALUE
$(-3)^2$	$(-3)(-3)$	+	9
$(-3)^3$	$(-3)(-3)(-3)$	−	-27
$(-2)^4$	four factors	+	16
-3^2	$-(3 \times 3)$	−	-9

$(-3)^2$ AND -3^2 ARE DIFFERENT

The brackets decide what is squared. Without them only the 3 is squared and the minus stays outside: $-3^2 = -9$. This distinction is tested in every year from now on.

02 Worked examples

EXAMPLE 1 **Compute $(-4) \times (-5)$, $4 \times (-5)$ and $(-2)(-3)(-4)$.**

1. Alike signs: positive. $4 \times 5 = 20$.
2. Different signs: negative. -20.
3. Three negatives — odd, so negative.
4. $2 \times 3 \times 4 = 24$, so -24.

ANSWER

$20, -20, -24$

EXAMPLE 2 **Compute $(-3)^2$, $(-3)^3$ and -3^2.**

1. $(-3)^2 = (-3)(-3) = 9$

   Two negatives.

2. $(-3)^3 = 9 \times (-3) = -27$

   Three negatives.

3. -3^2 means $-(3^2)$.

   No brackets.

4. $= -9$

ANSWER

$9, -27, -9$

EXAMPLE 3 **Find the sign of $(-5)(2)(-3)(-1)(4)$ without computing it.**

1. Count the negative factors: $-5, -3, -1$.
2. Three of them — an odd number.
3. The product is negative.
4. Its value is -120.

ANSWER

Negative; -120

03 Interactive model

Write the descending pattern (−4)×3, (−4)×2, … on the board and let the class continue it past zero; they produce the rule themselves.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Multiplication	Ko‘paytirish	Умножение
Product	Ko‘paytma	Произведение
Factor	Ko‘paytuvchi	Множитель
Sign rule	Ishoralar qoidasi	Правило знаков
Power	Daraja	Степень
Even number of factors	Juft sondagi ko‘paytuvchilar	Чётное число множителей
Square	Kvadrat	Квадрат
Cube	Kub	Куб

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(-4)(-5)$ equals:

- 20
- -20
- 9
- -9

$4 \times (-5)$ equals:

- 20
- -20
- -1
- 1

A product of three negatives is:

- positive
- negative
- zero
- undefined

$(-3)^2$ equals:

- 9
- -9
- 6
- -6

-3^2 equals:

- 9
- -9
- 6
- -6

A product containing a zero is:

- positive
- negative
- zero
- it depends

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4×5

   ANSWER 20

2. $4 \times (-5)$

   ANSWER -20

3. $(-4) \times 5$

   ANSWER -20

4. $(-4) \times (-5)$

   ANSWER 20

5. $(-7) \times 0$

   ANSWER 0

6. $(-3)^2$

   ANSWER 9

7. $(-3)^3$

   ANSWER -27

Medium · the standard question

7 PROBLEMS

1. $(-2)(-3)(-4)$

ANSWER -24

2. $(-2)(-3)(4)$

ANSWER 24

3. $(-1)^4$

ANSWER 1

4. $(-1)^5$

ANSWER -1

5. -3^2

ANSWER -9

6. $(-2)^4$

ANSWER 16

7. $(-5)(2)(-3)(-1)$

ANSWER -30

Hard · stretch and reason

7 PROBLEMS

1. $(-2)^3 \times (-3)^2$

   ANSWER -72

2. $(-1)^{100}$

   ANSWER 1

3. $(-1)^{101}$

   ANSWER -1

4. $(-2)^2 - (-2)^3$

   ANSWER 12

5. The sign of $(-1)(-2)(-3)\ldots(-9)$

   ANSWER Negative

6. Find x: $-3x = 21$

   ANSWER -7

7. $(-4)^2 \div (-2)^3$

   ANSWER -2

07 Homework

Homework — 5 tasks

Write the sign of every answer before computing its size.

1. Compute $(-6)(-7)$, $6 \times (-7)$ and $(-6)(7)$.
2. Compute $(-2)(-5)(-3)$ and $(-2)(-5)(3)$.
3. Compute $(-4)^2$, $(-4)^3$ and -4^2.
4. Find the sign of $(-1)(2)(-3)(4)(-5)$ without computing it.
5. Find x if $-5x = 35$.

Home / Grade 7 / Geometry / Quarter 1

Division of integers

The same sign rules as multiplication — and the one thing that is never allowed.

LESSON 31–32 | 2 × 40 MIN | MATEMATIKA 7, §13 | S8 1.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The sign rules	13 min
Division by zero	23 min
Order of operations	22 min
When is it an integer?	18 min
Homework	4 min

LEARNING OBJECTIVES

- Divide integers using the sign rules.
- Explain why division by zero is undefined.
- Decide whether a quotient of two integers is an integer.
- Use the order of operations with signed numbers.

TEXTBOOK

National	pp. 83–88
Cambridge	Stage 8, pp. 6–10
Workbook	Exercise 1.2

01 Explanation

The sign rules

Division is the reverse of multiplication, so the signs behave in exactly the same way.

SIGNS	QUOTIENT	EXAMPLE
$(+) \div (+)$	$+$	$20 \div 5 = 4$
$(+) \div (-)$	$-$	$20 \div (-5) = -4$
$(-) \div (+)$	$-$	$(-20) \div 5 = -4$
$(-) \div (-)$	$+$	$(-20) \div (-5) = 4$

Why. $(-20) \div (-5) = 4$ because $4 \times (-5) = -20$. Every division is checked by the multiplication that undoes it.

CHECK EVERY DIVISION BY MULTIPLYING BACK

It takes two seconds and catches both a wrong size and a wrong sign. It is the only check that needs nothing but the answer you already have.

Division by zero

> $a \div 0$ is undefined for every a

Why. $20 \div 0$ would be a number q with $q \times 0 = 20$ — but every product with zero is zero, so no such q exists.

EXPRESSION	VALUE
$0 \div 7$	0
$7 \div 0$	undefined
$0 \div 0$	undefined

> $0 \div 7$ AND $7 \div 0$ ARE COMPLETELY DIFFERENT
>
> The first is 0; the second does not exist. Zero on the top is fine, zero on the bottom never is — a rule that will matter in every algebra lesson from now on.

Order of operations

ORDER	OPERATION
1	brackets
2	powers
3	multiplication and division, left to right
4	addition and subtraction, left to right

Example. $-12 \div 4 + (-3) \times 2$. Divide and multiply first: $-3 + (-6)$. Then add: -9.

> LEFT TO RIGHT WITHIN THE SAME LEVEL
>
> $24 \div 4 \times 2$ is 12, not 3: division and multiplication have equal rank, so they are done in the order written.

When is it an integer?

A quotient of integers is an integer only when the divisor is a factor of the dividend.

DIVISION	INTEGER?	VALUE
$(-36) \div 9$	yes	-4
$(-36) \div (-12)$	yes	3
$(-36) \div 5$	no	-7.2
$7 \div (-2)$	no	-3.5

This is why $\mathbb{Z}$ is not closed under division, and why the next chapter introduces the rational numbers.

THE SIGN HAS NOTHING TO DO WITH IT

Whether the answer is a whole number depends only on the moduli. The sign is decided separately, and never affects divisibility.

02 Worked examples

EXAMPLE 1 **Compute $(-20) \div (-5)$, $20 \div (-5)$ and $(-36) \div 9$.**

1. Alike signs: positive. $20 \div 5 = 4$.
2. Different signs: negative. -4.
3. Different signs: $36 \div 9 = 4$, so -4.
4. Check: $(-4) \times 9 = -36$ ✓

ANSWER

$4, -4, -4$

EXAMPLE 2 **Compute $-12 \div 4 + (-3) \times 2$.**

1. Division and multiplication first.
2. $-12 \div 4 = -3$ and $(-3) \times 2 = -6$.
3. $-3 + (-6)$
4. $= -9$

ANSWER

-9

EXAMPLE 3 **Explain why $7 \div 0$ has no value, while $0 \div 7 = 0$.**

1. $7 \div 0 = q$ would mean $q \times 0 = 7$.
2. Every product with zero is zero, never 7.
   So no q exists.
3. $0 \div 7 = q$ means $q \times 7 = 0$.
4. $q = 0$ works — and only that.

ANSWER

Undefined; and 0

03 Interactive model

Ask how many times zero fits into seven; the class runs out of patience before the answer, which is exactly the point.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Division	Bo‘lish	Деление
Quotient	Bo‘linma	Частное
Dividend	Bo‘linuvchi	Делимое
Divisor	Bo‘luvchi	Делитель
Undefined	Aniqlanmagan	Не определено
Order of operations	Amallar tartibi	Порядок действий
Exactly divisible	Butun bo‘linadi	Делится нацело
Remainder	Qoldiq	Остаток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(-20) \div (-5)$ equals:

- 4
- -4
- 100
- -100

$20 \div (-5)$ equals:

- 4
- -4
- 15
- -15

$0 \div 7$ equals:

- 0
- 7
- undefined
- 1

$7 \div 0$ is:

- 0
- 7
- undefined
- ∞

$24 \div 4 \times 2$ equals:

- 3
- 12
- 48
- 192

$(-36) \div 5$ is:

- an integer
- not an integer
- zero
- undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $20 \div 5$

   ANSWER 4

2. $20 \div (-5)$

   ANSWER -4

3. $(-20) \div 5$

   ANSWER -4

4. $(-20) \div (-5)$

   ANSWER 4

5. $0 \div 9$

   ANSWER 0

6. $9 \div 0$

   ANSWER Undefined

7. $(-36) \div 9$

   ANSWER -4

Medium · the standard question

7 PROBLEMS

1. $(-36) \div (-12)$

ANSWER 3

2. $(-45) \div 9$

ANSWER -5

3. $-12 \div 4 + (-3) \times 2$

ANSWER -9

4. $24 \div 4 \times 2$

ANSWER 12

5. $(-18) \div (-3) - 4$

ANSWER 2

6. $(-2)^3 \div 4$

ANSWER -2

7. Is $(-36) \div 5$ an integer?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $(-48) \div (-6) \div (-2)$

   ANSWER -4

2. $(-3)^2 \times 2 - 30 \div (-5)$

   ANSWER 24

3. $(-60) \div (2 - 5)$

   ANSWER 20

4. Find x: $\frac{x}{-4} = 7$

   ANSWER -28

5. Find x: $\frac{-36}{x} = -9$

   ANSWER 4

6. For which integers n is $\frac{12}{n}$ an integer?

   ANSWER $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

7. Why is $0 \div 0$ undefined?

   ANSWER Every number would satisfy it

07 Homework

Homework — 5 tasks

Check every division by multiplying the answer back.

1. Compute $(-42) \div 7$, $42 \div (-7)$ and $(-42) \div (-7)$.
2. Compute $-20 \div 5 + (-4) \times 3$.
3. Compute $(-2)^4 \div (-8)$.
4. Explain why $5 \div 0$ is undefined.

5. List all integers n for which $\frac{18}{n}$ is an integer.

Problems on the four operations with integers

Mixed calculations and word problems — where the sign rules meet the real world.

LESSON 33–34 | 2 × 40 MIN | MATEMATIKA 7, §14 | S8 1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Mixed expressions	13 min
The distributive law	23 min
Word problems	22 min
Averages	18 min
Homework	4 min

LEARNING OBJECTIVES

- Evaluate expressions containing all four operations and brackets.
- Translate a worded situation into an expression with signs.
- Use the distributive law with negative numbers.
- Check an answer for plausibility as well as for arithmetic.

TEXTBOOK

National	pp. 89–94
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1 review

01 Explanation

Mixed expressions

Work brackets first, then powers, then × and ÷, then + and −.

EXPRESSION	WORKING	VALUE
$-5 + 3 \times (-4)$	$-5 + (-12)$	-17
$(-5 + 3) \times (-4)$	$(-2)(-4)$	8
$-20 \div (-4) - 6$	$5 - 6$	-1
$(-3)^2 - 4 \times (-2)$	$9 + 8$	17

THE BRACKETS CHANGE EVERYTHING

The first two rows use the same three numbers and give -17 and 8. Copying an expression without its brackets is the most expensive mistake in this topic.

The distributive law

$$a(b + c) = ab + ac$$

It holds for negative numbers exactly as for positive ones, and it is how brackets are removed.

EXPRESSION	EXPANDED	VALUE
$-3(4 + 5)$	$-12 - 15$	-27
$-3(4 - 5)$	$-12 + 15$	3
$-(x - 7)$	$-x + 7$	—
$5 - (3 - 8)$	$5 - 3 + 8$	10

A MINUS IN FRONT OF A BRACKET CHANGES **EVERY** SIGN INSIDE

$-(x - 7) = -x + 7$, not $-x - 7$. Missing the second sign is the classic slip, and it will recur in every algebra lesson this year.

Word problems

SITUATION	EXPRESSION	ANSWER
$-7°$, rising $3°$ an hour for 4 hours	$-7 + 3 \times 4$	$5°$
a debt of $20\,000$, paid off at 6000 a month for 3 months	$-20\,000 + 3 \times 6000$	-2000
a diver at $-18\ m$ descending $4\ m$ a minute for 5 minutes	$-18 - 4 \times 5$	$-38\ m$
a lift on floor 12 going down 3 floors 4 times	$12 - 3 \times 4$	0

WRITE THE EXPRESSION BEFORE COMPUTING ANYTHING

The mathematics of these problems is easy; the translation is not. Getting the expression on paper first separates the two difficulties.

Averages

The **average** (arithmetic mean) of a list is the sum divided by how many there are.
With negatives it works exactly the same way.

$$mean = \frac{\text{sum of the values}}{\text{how many values}}$$

LIST	SUM	COUNT	MEAN
$-3, 0, 2, -5, 1$	-5	5	-1
$-8, -4, 0, 4$	-8	4	-2
$-6, 6$	0	2	0

THE MEAN ALWAYS LIES BETWEEN THE LEAST AND THE GREATEST

A mean of -1 for a list running from -5 to 2 is plausible; a mean of 3 would not be. That single check catches most arithmetic slips.

02 Worked examples

EXAMPLE 1 **Evaluate $-5 + 3 \times (-4)$ and $(-5 + 3) \times (-4)$.**

1. First: multiply before adding.
2. $-5 + (-12) = -17$
3. Second: bracket first.
4. $(-2)(-4) = 8$

   Same numbers, different answers.

ANSWER

-17 and 8

EXAMPLE 2 **The temperature is $-7°C$ and rises by 3 degrees each hour for 4 hours. Find it then.**

1. Expression: $-7 + 3 \times 4$.
2. $3 \times 4 = 12$
3. $-7 + 12$
4. $= 5°C$

   Now above zero.

ANSWER

$5°C$

EXAMPLE 3 **Find the mean of $-3, 0, 2, -5, 1$.**

1. Sum: $-3 + 0 + 2 - 5 + 1$.
2. $= -5$
3. Five values: $-5 \div 5$.
4. $= -1$ — between -5 and 2 ✓

ANSWER

-1

03 Interactive model

Give the same three numbers with and without brackets and let the class compute both; the difference makes the order of operations matter rather than merely exist.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Expression	Ifoda	Выражение
Brackets	Qavslar	Скобки
Distributive law	Taqsimot qonuni	Распределительный закон
To evaluate	Qiymatini topmoq	Вычислить
Word problem	Matnli masala	Текстовая задача
Average	O‘rta qiymat	Среднее
Net change	Sof o‘zgarish	Чистое изменение
Plausible	Ishonarli	Правдоподобный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$-5 + 3 \times (-4)$ equals:

- 8
- -17
- -32
- 32

$(-5 + 3) \times (-4)$ equals:

- 8
- -17
- -8
- 2

$-(x - 7)$ equals:

- $-x - 7$
- $-x + 7$
- $x - 7$
- $x + 7$

$5 - (3 - 8)$ equals:

- 0
- 10
- -6
- 16

The mean of $-8, -4, 0, 4$:

- -2
- -8
- 0
- 2

A mean always lies:

- above the greatest
- below the least
- between them
- at zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $-5 + 3 \times (-4)$

   ANSWER -17

2. $(-5 + 3) \times (-4)$

   ANSWER 8

3. $-20 \div (-4) - 6$

   ANSWER -1

4. $(-3)^2 - 4 \times (-2)$

   ANSWER 17

5. $-3(4 + 5)$

   ANSWER -27

6. $-3(4 - 5)$

   ANSWER 3

7. $5 - (3 - 8)$

   ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Mean of $-3, 0, 2, -5, 1$

   ANSWER -1

2. Mean of $-8, -4, 0, 4$

   ANSWER -2

3. $-7 + 3 \times 4$

   ANSWER 5

4. $-18 - 4 \times 5$

   ANSWER -38

5. $-20\,000 + 3 \times 6000$

   ANSWER -2000

6. Simplify $-(a - b + c)$

   ANSWER $-a + b - c$

7. $(-2)(-3) - (-4)(5)$

   ANSWER 26

Hard · stretch and reason

7 PROBLEMS

1. $(-4)^2 - (-4)^2 \div (-2)^3$

   ANSWER 18

2. $-2[3 - (-5 + 1)]$

   ANSWER -14

3. Mean of $-12, -5, 0, 3, 9$

   ANSWER -1

4. A diver at $-12\ m$ rises $2\ m$ a minute for 8 minutes

   ANSWER $+4\ m$

5. Five days at $-4, -1, 2, -7, 5$: the mean temperature

   ANSWER $-1°$

6. Find x: $3x - (-6) = 0$

   ANSWER -2

7. Evaluate $a^2 - 2ab$ at $a = -3, b = 4$

   ANSWER 33

07 Homework

Homework — 5 tasks

Write the expression before the arithmetic, and check the answer for plausibility.

1. Evaluate $-6 + 4 \times (-3)$ and $(-6 + 4) \times (-3)$.
2. Evaluate $(-2)^3 + 5 \times (-4) \div 2$.
3. Simplify $-(2x - 5 + y)$.
4. The temperature is $-9°C$ and rises 4 degrees each hour for 3 hours. Find it then.
5. Find the mean of $-10, -2, 0, 4, 8$.

Control work 2, and work on the mistakes

The integers in one paper — the four operations, the signs and the order.

LESSON 35–36 | 2 × 40 MIN | MATEMATIKA 7, NAZORAT ISHI 2 | S8 1 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Use the sign rules for all four operations under time.
- Apply the order of operations to a mixed expression.
- Compare and order integers, and use the modulus.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 36–94
Cambridge	Stage 8, pp. 2–10
Workbook	Control paper M2

01 Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Put in ascending order: $-7, 3, 0, -12, 5$	*4*	L23–24
2	Find $\lvert -9 \rvert$, $-(-6)$ and the opposite of 11	*4*	L21–22
3	Compute $(-8) + (-5)$, $12 + (-17)$, $-4 - (-9)$	*6*	L25–28
4	Compute $(-6)(-7)$, $(-45) \div 9$, $(-2)^4$	*6*	L29–32
5	Evaluate $-15 \div 3 + (-2) \times 4$	*5*	L33–34
6	Solve $\lvert x \rvert = 7$ and find the mean of $-5, 1, -2, 6$	*5*	L21–22, 33–34

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for putting the negatives in the right order among themselves; Q4 two for the even power; Q5 two for doing ÷ and × before + ; Q6 two for giving **both** solutions.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
negatives ordered forwards	$-7, -12$	$-12, -7$
modulus left negative	$\lvert -9 \rvert = -9$	9
same signs subtracted	$(-8) + (-5) = -3$	-13
sign taken from the first number	$12 + (-17) = 5$	-5
double minus missed	$-4 - (-9) = -13$	5
even power made negative	$(-2)^4 = -16$	16
left to right regardless	$-15 \div 3 + (-2) \times 4 = -28$	-13

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter II as one map

BLOCK	THE SENTENCE
why negatives exist	so that every subtraction has an answer
the coordinate line	further right means greater
the modulus	the distance from zero — never negative
addition	same signs add and keep; different signs subtract and take the larger sign
subtraction	$a - b = a + (-b)$
multiplication and division	alike gives +, unlike gives −
powers	even exponent positive, odd exponent keeps the sign
order	brackets, powers, ×÷, +−

LOOKING FORWARD

The next chapter fills the last gap in the number system for this year: the **rational** numbers, where division always works too. The sign rules carry over unchanged — only the arithmetic of fractions is added.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: $(-8) + (-5)$, $12 + (-17)$, $-4 - (-9)$.**

1. Same signs: $8 + 5 = 13$, keep the minus — -13.
2. Different signs: $17 - 12 = 5$, sign of -17 — -5.
3. Rewrite: $-4 + 9$.
   Two minuses.
4. $= 5$

ANSWER

$-13, -5, 5$

EXAMPLE 2 **Model answer, Q5: $-15 \div 3 + (-2) \times 4$.**

1. Division and multiplication first.
2. $-15 \div 3 = -5$
3. $(-2) \times 4 = -8$
4. $-5 + (-8) = -13$

ANSWER

-13

EXAMPLE 3 **Model answer, Q6: solve $|x| = 7$ and find the mean of $-5, 1, -2, 6$.**

1. Two points are 7 from zero.

2. $x = 7$ or $x = -7$.

   Both needed.

3. Sum: $-5 + 1 - 2 + 6 = 0$.

4. Mean $= 0 \div 4 = 0$.

ANSWER

$x = \pm 7$; mean 0

03 Interactive model

Mark Q3 aloud, reading each sum as a sentence about money; every wrong sign becomes obviously wrong before any rule is quoted.

Chapter II in twelve questions

Negatives exist so that:

- addition works
- subtraction always works
- division works
- nothing

Zero is:

- positive
- negative
- neither
- both

Greater: -8 or -3?

- -8
- -3
- equal
- cannot tell

$|-9|$ equals:

- -9
- 9
- 0
- 81

$(-8) + (-5)$ equals:

- -3
- -13
- 3
- 13

$12 + (-17)$ equals:

- 5
- -5
- 29
- -29

$a - b$ equals:

- $a + b$
- $a + (-b)$
- $b - a$
- $-a - b$

$-4 - (-9)$ equals:

- -13
- 5
- -5
- 13

$(-6)(-7)$ equals:

- 42
- -42
- 13
- -13

$(-45) \div 9$ equals:

- 5
- -5
- -405
- 405

$(-2)^4$ equals:

- 16
- -16
- 8
- -8

Order of operations:

- left to right always
- brackets, powers, ×÷, +−
- +− first
- no rule

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Integer	Butun son	Целое число
Modulus	Modul	Модуль
Sign rule	Ishoralar qoidasi	Правило знаков
Order of operations	Amallar tartibi	Порядок действий
Ascending order	O‘sish tartibi	По возрастанию
Opposite number	Qarama-qarshi son	Противоположное число
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 tests mainly:

- arithmetic
- ordering negatives
- speed
- the modulus

In Q3 the third part needs:

- the addition rule
- the double-minus rule
- a power
- a modulus

$(-2)^4$ is positive because:

- the base is even
- the exponent is even
- the base is negative
- no reason

Q5 requires:

- left to right
- × and ÷ first
- + and − first
- brackets

Q6’s equation has:

- one answer
- two answers
- no answer
- many

Work on the mistakes means:

- fix the wrong line
- rewrite the solution
- copy the answer
- skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Ascending: $-7, 3, 0, -12, 5$

   ANSWER $-12, -7, 0, 3, 5$

2. $|-9|$

   ANSWER 9

3. $-(-6)$

   ANSWER 6

4. Opposite of 11

   ANSWER -11

5. $(-8) + (-5)$

   ANSWER -13

6. $12 + (-17)$

   ANSWER -5

7. $-4 - (-9)$

   ANSWER 5

Medium · the standard question 7 PROBLEMS

1. $(-6)(-7)$

   ANSWER 42

2. $(-45) \div 9$

   ANSWER -5

3. $(-2)^4$

   ANSWER 16

4. $-15 \div 3 + (-2) \times 4$

   ANSWER -13

5. Solve $|x| = 7$

   ANSWER ± 7

6. Mean of $-5, 1, -2, 6$

   ANSWER 0

7. $(-3)^3$

   ANSWER -27

Hard · stretch and reason

7 PROBLEMS

1. $(-2)^3 + (-3)^2 - (-4)$

   ANSWER 5

2. $-24 \div (-6) \times (-2)$

   ANSWER -8

3. $(-5)(3) - (-2)(-6)$

   ANSWER -27

4. Mean of $-9, -3, 0, 6, 11$

   ANSWER 1

5. Solve $|x| \le 2$ in integers

   ANSWER $-2, -1, 0, 1, 2$

6. Find x: $-4x + 3 = 19$

   ANSWER -4

7. Evaluate $2a - b^2$ at $a = -5, b = -3$

   ANSWER -19

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the next chapter begins.

1. Put in ascending order: $-4, 8, 0, -15, 2$.
2. Compute $(-9) + (-6)$, $14 + (-21)$ and $-7 - (-12)$.
3. Compute $(-8)(-4)$, $(-56) \div 7$ and $(-3)^4$.
4. Evaluate $-18 \div 6 + (-3) \times 5$.
5. Solve $|x| = 12$ and find the mean of $-7, 2, -4, 9$.

The idea of the rational numbers

Every fraction, positive or negative — the set in which all four operations work.

LESSON 37–38 | 2 × 40 MIN | MATEMATIKA 7, §15 | S8 7.1

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The definition	13 min
Everything is a fraction	23 min
Fractions and decimals	22 min
On the line	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define a rational number and use the symbol $\mathbb{Q}$.
- Write an integer, a decimal and a mixed number as a fraction.
- Convert between fractions and decimals, including recurring ones.
- Place rational numbers on the coordinate line.

TEXTBOOK

National	pp. 95–100
Cambridge	Stage 8, pp. 66–72
Workbook	Exercise 7.1

01 Explanation

The definition

> A number is rational if it can be written as $\frac{p}{q}$ with $p, q \in \mathbb{Z}$ and $q \neq 0$

The set of them is $\mathbb{Q}$. It contains every integer, every fraction, every terminating decimal and every recurring decimal — positive and negative alike.

OPERATION	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$
addition	✓	✓	✓
subtraction	✗	✓	✓
multiplication	✓	✓	✓
division (not by 0)	✗	✗	✓

$\mathbb{Q}$ IS THE FIRST SET IN WHICH ALL FOUR OPERATIONS WORK

That is the whole reason for the chapter. Grade 8 will find that $\sqrt{2}$ is **not** rational, and the system will have to grow once more.

Everything is a fraction

NUMBER	AS $\frac{P}{Q}$
7	$\frac{7}{1}$
-4	$\frac{-4}{1}$
0.75	$\frac{3}{4}$
$2\frac{1}{3}$	$\frac{7}{3}$
-1.6	$-\frac{8}{5}$
0	$\frac{0}{1}$

A fraction may be written in many equivalent ways: $\frac{3}{4} = \frac{6}{8} = \frac{75}{100}$. The **simplest form** has no common factor above 1 in numerator and denominator.

THE DENOMINATOR MAY NEVER BE ZERO

$\frac{5}{0}$ is not a rational number; it is not a number at all. The condition $q \neq 0$ is part of the definition, not an afterthought.

Fractions and decimals

Divide the numerator by the denominator. The result is either terminating or recurring — never anything else.

FRACTION	DECIMAL	KIND
$\frac{3}{4}$	0.75	terminating
$\frac{1}{8}$	0.125	terminating
$\frac{1}{3}$	$0.333\ldots$	recurring
$\frac{2}{11}$	$0.1818\ldots$	recurring
$\frac{5}{6}$	$0.8333\ldots$	recurring

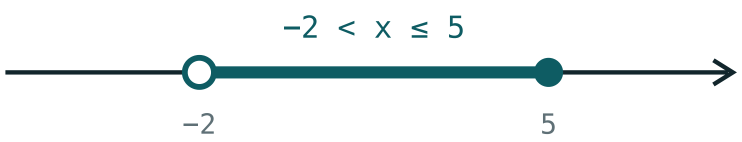
A fraction in simplest form terminates exactly when its denominator has no prime factors other than 2 and 5 — because $10 = 2 \times 5$.

RATIONAL ⟺ TERMINATING OR RECURRING

A decimal that neither stops nor repeats — such as $0.101001000100001\ldots$ — is not rational. Grade 8 will meet a famous one: $\sqrt{2}$.

On the line

Every rational number has a place. To mark $\frac{3}{4}$, divide the unit into 4 parts and take 3; to mark $-\frac{3}{4}$, do the same to the left.



open circle = strict · filled = included

Dividing the unit segment places any fraction exactly.

NUMBER	LIES BETWEEN
$\frac{3}{4}$	*0 and 1*
$-\frac{3}{4}$	*−1 and 0*
$\frac{7}{3}$	*2 and 3*
$-\frac{7}{3}$	*−3 and −2*

BETWEEN ANY TWO RATIONALS THERE IS ANOTHER

Their average. So the rationals leave no gaps a ruler can see — and yet Grade 8 will show that gaps remain.

02 Worked examples

EXAMPLE 1 **Write 7, -1.6 and $2\frac{1}{3}$ as fractions $\frac{p}{q}$.**

1. $7 = \frac{7}{1}$

   Every integer is a fraction.

2. $-1.6 = -\frac{16}{10} = -\frac{8}{5}$

   Simplify.

3. $2\frac{1}{3} = \frac{6+1}{3}$

4. $= \frac{7}{3}$

ANSWER

$\frac{7}{1}, -\frac{8}{5}, \frac{7}{3}$

EXAMPLE 2 **Convert $\frac{3}{8}$ and $\frac{5}{6}$ to decimals, and say which recurs.**

1. $3 \div 8 = 0.375$

   Terminating.

2. $8 = 2^3$ — only twos.

3. $5 \div 6 = 0.8333\ldots$

   Recurring.

4. $6 = 2 \times 3$ — the 3 causes it.

ANSWER

0.375 and $0.8333\ldots$

EXAMPLE 3 **Between which two integers does $-\frac{7}{3}$ lie?**

1. $\frac{7}{3} = 2\frac{1}{3}$

2. So $-\frac{7}{3} = -2\frac{1}{3}$.

3. That is further left than -2.

4. Between -3 and -2.

ANSWER

-3 and -2

03 Interactive model

Ask for a number between 0.33 and 0.34, then between that and 0.34; the class discovers that the process never ends, which is what “dense” means.

Fractions and their simplest form

$$\frac{18}{24}$$

1. $18 = 2 \cdot 3^2$ and $24 = 2^3 \cdot 3$

Factorise both.

2. The common factor is 6.

The highest one — anything less leaves work to do.

3. $$\frac{18 \div 6}{24 \div 6} = \frac{3}{4}$$

Divide both parts by it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational number	Ratsional son	Рациональное число
Fraction	Kasr	Дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Terminating decimal	Chekli o‘nli kasr	Конечная десятичная дробь
Recurring decimal	Davriy o‘nli kasr	Периодическая дробь
Mixed number	Aralash son	Смешанное число
Equivalent fractions	Teng kasrlar	Равные дроби

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rational number is:

- any decimal
- $\frac{p}{q}$ with integers, $q \neq 0$
- an integer
- a fraction below 1

Is 7 rational?

- yes
- no
- sometimes
- undefined

$\mathbb{Q}$ is closed under:

- addition only
- all four operations (not ÷0)
- subtraction only
- none

$\frac{3}{8}$ as a decimal:

- 0.38
- 0.375
- 0.83
- 0.125

A fraction terminates when its denominator has only:

- 2 and 3
- 2 and 5
- 3 and 5
- primes

$-\frac{7}{3}$ lies between:

- -2 and -1
- -3 and -2
- 2 and 3
- -1 and 0

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. 7 as a fraction

ANSWER $\frac{7}{1}$

2. -4 as a fraction

ANSWER $-\frac{4}{1}$

3. 0.75 as a fraction

ANSWER $\frac{3}{4}$

4. $2\frac{1}{3}$ as a fraction

ANSWER $\frac{7}{3}$

5. $\frac{1}{4}$ as a decimal

ANSWER 0.25

6. $\frac{1}{3}$ as a decimal

ANSWER $0.333\ldots$

7. Is $\frac{5}{0}$ a number?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. -1.6 as a fraction

ANSWER $-\dfrac{8}{5}$

2. $\dfrac{3}{8}$ as a decimal

ANSWER 0.375

3. $\dfrac{5}{6}$ as a decimal

ANSWER $0.8333\ldots$

4. $\dfrac{2}{11}$ as a decimal

ANSWER $0.1818\ldots$

5. Does $\dfrac{7}{20}$ terminate?

ANSWER Yes

6. Does $\dfrac{7}{30}$ terminate?

ANSWER No

7. $-\dfrac{7}{3}$ lies between

ANSWER -3 and -2

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\dfrac{84}{126}$

ANSWER $\dfrac{2}{3}$

2. 0.375 as a fraction in lowest terms

ANSWER $\dfrac{3}{8}$

3. A number between $\dfrac{1}{3}$ and $\dfrac{1}{2}$

ANSWER $\dfrac{5}{12}$

4. Order $\dfrac{2}{3}, \dfrac{3}{5}, \dfrac{5}{8}$

ANSWER $\dfrac{3}{5} < \dfrac{5}{8} < \dfrac{2}{3}$

5. Which of $\dfrac{1}{6}, \dfrac{3}{16}, \dfrac{9}{25}$ terminate?

ANSWER $\dfrac{3}{16}$ and $\dfrac{9}{25}$

6. Is every integer rational?

ANSWER Yes

7. Is every rational an integer?

ANSWER No

07 Homework

Homework — 5 tasks

Give every fraction in its simplest form.

1. Write 9, -2.4 and $3\frac{2}{5}$ as fractions.
2. Convert $\frac{5}{8}$, $\frac{4}{9}$ and $\frac{7}{20}$ to decimals.
3. Say which of them terminate, and why.
4. Simplify $\frac{72}{108}$.
5. Find a rational number between $\frac{2}{5}$ and $\frac{1}{2}$.

Addition of rational numbers

A common denominator, then the integer sign rules — nothing else is new.

LESSON 39–40 | 2 × 40 MIN | MATEMATIKA 7, §16 | S8 7.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Same denominator	13 min
Different denominators	23 min
With signs	22 min
Mixed numbers	18 min
Homework	4 min

LEARNING OBJECTIVES

- Add fractions with the same and with different denominators.
- Add positive and negative fractions using the sign rules.
- Add mixed numbers and decimals.
- Use the commutative and associative laws to simplify a sum.

TEXTBOOK

National	pp. 101–107
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01 Explanation

Same denominator

$$\frac{a}{c} + \frac{b}{c} = \frac{a + b}{c}$$

Add the numerators and keep the denominator; then simplify.

SUM	WORKING	ANSWER
$\frac{2}{7} + \frac{3}{7}$	$\frac{5}{7}$	$\frac{5}{7}$
$\frac{3}{8} + \frac{1}{8}$	$\frac{4}{8}$	$\frac{1}{2}$
$-\frac{5}{9} + \frac{2}{9}$	$\frac{-3}{9}$	$-\frac{1}{3}$

DENOMINATORS ARE NOT ADDED

> $\frac{2}{7} + \frac{3}{7}$ is $\frac{5}{7}$, not $\frac{5}{14}$. Thinking of sevenths as a unit — two sevenths plus three sevenths — makes the rule obvious.

Different denominators

STEP	WHAT TO DO
1	find the LCM of the denominators
2	rewrite each fraction with that denominator
3	add the numerators
4	simplify

Example. $\frac{3}{4} + \frac{5}{6}$. LCM of 4 and 6 is 12:

$$\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$$

> USE THE LCM, NOT THE PRODUCT
>
> $4 \times 6 = 24$ also works, but leaves $\frac{38}{24}$ to simplify. The LCM gives the shortest route and the least cancelling.

With signs

Once the denominators agree, the numerators are added exactly as integers were in the last chapter.

SUM	COMMON DENOMINATOR	ANSWER
$-\frac{1}{2} + \frac{1}{3}$	$-\frac{3}{6} + \frac{2}{6}$	$-\frac{1}{6}$
$-\frac{3}{4} + (-\frac{1}{6})$	$-\frac{9}{12} - \frac{2}{12}$	$-\frac{11}{12}$
$\frac{5}{8} + (-\frac{5}{8})$	$\frac{0}{8}$	0

TWO CHAPTERS, ONE CALCULATION

Chapter II supplied the sign rules; this chapter supplies the common denominator. Every question here is one of each, done in that order.

Mixed numbers

Two routes: convert to improper fractions, or add whole parts and fractional parts separately.

SUM	ROUTE A	ROUTE B
$2\frac{1}{3} + 1\frac{1}{2}$	$\frac{7}{3} + \frac{3}{2} = \frac{23}{6}$	$3 + \frac{5}{6}$

Both give $3\frac{5}{6}$. Route B is quicker when the whole parts are large; Route A is safer when signs are involved.

WITH NEGATIVE MIXED NUMBERS, CONVERT FIRST

$-2\frac{1}{3}$ means $-(2 + \frac{1}{3}) = -\frac{7}{3}$, not $-2 + \frac{1}{3}$. Route A removes the ambiguity entirely.

02 Worked examples

EXAMPLE 1 **Compute $\frac{3}{4} + \frac{5}{6}$.**

1. LCM of 4 and 6 is 12.
2. $\frac{9}{12} + \frac{10}{12}$
3. $= \frac{19}{12}$
4. $= 1\frac{7}{12}$

ANSWER

$1\frac{7}{12}$

EXAMPLE 2 **Compute $-\frac{1}{2} + \frac{1}{3}$ and $-\frac{3}{4} + \left(-\frac{1}{6}\right)$.**

1. $-\frac{3}{6} + \frac{2}{6} = -\frac{1}{6}$

   Different signs.

2. LCM of 4 and 6 is 12.

3. $-\frac{9}{12} - \frac{2}{12}$

   Same signs.

4. $= -\frac{11}{12}$

ANSWER

$-\frac{1}{6}$ and $-\frac{11}{12}$

EXAMPLE 3 **Compute $2\frac{1}{3} + 1\frac{1}{2}$.**

1. Whole parts: $2 + 1 = 3$.

   Route B.

2. Fractions: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$

3. $3 + \frac{5}{6}$

4. $= 3\frac{5}{6}$

ANSWER

$3\frac{5}{6}$

03 Interactive model

Cut two paper strips into quarters and sixths; the class sees that twelfths are the smallest pieces both can be made from, and the LCM stops being a rule.

Finding the common denominator

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
6	$2 \cdot 3$	4
8	2^3	3

LOWEST COMMON DENOMINATOR

$$24$$

$$\frac{20}{24} + \frac{9}{24} = \frac{29}{24}.$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
To simplify	Qisqartirish	Сократить
Mixed number	Aralash son	Смешанное число
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Sum	Yig‘indi	Сумма
Like denominators	Bir xil maxrajlar	Одинаковые знаменатели
LCM	EKUK	НОК

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{2}{7} + \frac{3}{7}$ equals:

- $\frac{5}{14}$
- $\frac{5}{7}$
- $\frac{6}{7}$
- $\frac{6}{49}$

For different denominators, first find:

- the product
- the LCM
- the HCF
- the sum

$\frac{3}{4} + \frac{5}{6}$ equals:

- $\frac{8}{10}$
- $1\frac{7}{12}$
- $\frac{19}{24}$
- 2

$-\frac{1}{2} + \frac{1}{3}$ equals:

- $\frac{1}{6}$
- $-\frac{1}{6}$
- $-\frac{5}{6}$
- $\frac{5}{6}$

$-2\frac{1}{3}$ as an improper fraction:

- $-\frac{7}{3}$
- $-\frac{5}{3}$
- $\frac{7}{3}$
- $-\frac{1}{3}$

$\frac{5}{8} + (-\frac{5}{8})$ equals:

- $\frac{10}{8}$
- 0
- $\frac{5}{4}$
- $-\frac{5}{8}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\dfrac{2}{7} + \dfrac{3}{7}$

ANSWER $\dfrac{5}{7}$

2. $\dfrac{3}{8} + \dfrac{1}{8}$

ANSWER $\dfrac{1}{2}$

3. $\dfrac{1}{2} + \dfrac{1}{4}$

ANSWER $\dfrac{3}{4}$

4. $\dfrac{1}{3} + \dfrac{1}{6}$

ANSWER $\dfrac{1}{2}$

5. $-\dfrac{5}{9} + \dfrac{2}{9}$

ANSWER $-\dfrac{1}{3}$

6. $\dfrac{5}{8} + (-\dfrac{5}{8})$

ANSWER 0

7. $0.4 + 0.35$

ANSWER 0.75

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{4} + \frac{5}{6}$

ANSWER $1\frac{7}{12}$

2. $-\frac{1}{2} + \frac{1}{3}$

ANSWER $-\frac{1}{6}$

3. $-\frac{3}{4} + (-\frac{1}{6})$

ANSWER $-\frac{11}{12}$

4. $2\frac{1}{3} + 1\frac{1}{2}$

ANSWER $3\frac{5}{6}$

5. $\frac{2}{3} + \frac{3}{5}$

ANSWER $1\frac{4}{15}$

6. $-\frac{5}{6} + \frac{3}{4}$

ANSWER $-\frac{1}{12}$

7. $1\frac{1}{4} + (-\frac{1}{2})$

ANSWER $\frac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

ANSWER 1

2. $-\frac{2}{3} + \frac{3}{4} - \frac{1}{2}$

ANSWER $-\frac{5}{12}$

3. $-1\frac{2}{5} + 2\frac{3}{10}$

ANSWER $\frac{9}{10}$

4. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

ANSWER $\frac{15}{16}$

5. Find x: $x + \frac{2}{3} = \frac{1}{6}$

ANSWER $-\frac{1}{2}$

6. $0.25 + \frac{1}{3}$

ANSWER $\frac{7}{12}$

7. The mean of $\frac{1}{2}$ and $-\frac{1}{6}$

ANSWER $\frac{1}{6}$

07 Homework

Homework — 5 tasks

Find the LCM before writing anything, and simplify every answer.

1. Compute $\frac{2}{5} + \frac{1}{3}$ and $\frac{5}{6} + \frac{3}{8}$.
2. Compute $-\frac{2}{3} + \frac{1}{4}$.
3. Compute $-\frac{1}{5} + (-\frac{3}{10})$.
4. Compute $3\frac{1}{4} + 2\frac{2}{3}$.
5. Find x if $x + \frac{3}{4} = \frac{1}{2}$.

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Subtraction of rational numbers

Add the opposite, exactly as with integers — the fractions change nothing.

LESSON 41–42 | 2 × 40 MIN | MATEMATIKA 7, §17 | S8 7.2

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

The rule	13 min
Different denominators	23 min
Mixed numbers	22 min
Differences on the line	18 min
Homework	4 min

LEARNING OBJECTIVES

- Subtract fractions with the same and with different denominators.
- Use a − b = a + (−b) for rational numbers.
- Subtract mixed numbers, including with borrowing.
- Find the difference between two rational numbers on the line.

TEXTBOOK

National	pp. 108–113
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01 Explanation

The rule

$$a - b = a + (-b)$$

Exactly as for integers. With like denominators, subtract the numerators:

$$\frac{a}{c} - \frac{b}{c} = \frac{a - b}{c}$$

DIFFERENCE	WORKING	ANSWER
$\frac{5}{7} - \frac{2}{7}$	$\frac{3}{7}$	$\frac{3}{7}$
$\frac{2}{7} - \frac{5}{7}$	$\frac{-3}{7}$	$-\frac{3}{7}$
$\frac{1}{4} - (-\frac{1}{4})$	$\frac{1}{4} + \frac{1}{4}$	$\frac{1}{2}$

TWO MINUSES GIVE A PLUS HERE TOO

$\frac{1}{4} - (-\frac{1}{4})$ is $\frac{1}{2}$. The rules of Chapter II apply to every rational number, not only to integers.

Different denominators

Same four steps as for addition: LCM, rewrite, subtract, simplify.

Example. $\frac{5}{6} - \frac{3}{4}$. LCM is *12*:

$$\frac{10}{12} - \frac{9}{12} = \frac{1}{12}$$

DIFFERENCE	COMMON DENOMINATOR	ANSWER
$\frac{3}{4} - \frac{1}{6}$	$\frac{9}{12} - \frac{2}{12}$	$\frac{7}{12}$
$\frac{1}{3} - \frac{1}{2}$	$\frac{2}{6} - \frac{3}{6}$	$-\frac{1}{6}$
$-\frac{2}{5} - \frac{3}{10}$	$-\frac{4}{10} - \frac{3}{10}$	$-\frac{7}{10}$

A NEGATIVE ANSWER IS NOT A MISTAKE

$\frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$ because a third is less than a half. Estimating which is bigger before subtracting tells you the sign in advance.

Mixed numbers

Two routes, as for addition. Route A — convert to improper fractions — never fails.

Example. $4\frac{1}{5} - 1\frac{3}{5}$.

ROUTE	WORKING
A	$\frac{21}{5} - \frac{8}{5} = \frac{13}{5} = 2\frac{3}{5}$
B	$4\frac{1}{5} = 3\frac{6}{5}$

Route B needs **borrowing**: since $\frac{1}{5}$ is smaller than $\frac{3}{5}$, one whole is exchanged for $\frac{5}{5}$. Both routes give $2\frac{3}{5}$.

BORROWING IS WHERE THE ERRORS ARE

$4\frac{1}{5}$ becomes $3\frac{6}{5}$, not $3\frac{11}{5}$. Converting to improper fractions avoids the step entirely, and is worth the extra line.

Differences on the line

The distance between two rational numbers is $|a - b|$, exactly as for integers.

POINTS	DIFFERENCE	DISTANCE
$\frac{3}{4}, \frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{2}$
$-\frac{1}{2}, \frac{1}{3}$	$-\frac{5}{6}$	$\frac{5}{6}$
$-\frac{3}{4}, -\frac{1}{4}$	$-\frac{1}{2}$	$\frac{1}{2}$

WHICH IS LARGER? SUBTRACT AND LOOK AT THE SIGN

A positive difference means the first is larger; a negative one means the second is. That is often quicker than finding a common denominator twice.

02 Worked examples

EXAMPLE 1 **Compute $\frac{5}{6} - \frac{3}{4}$.**

1. LCM of 6 and 4 is 12.
2. $\frac{10}{12} - \frac{9}{12}$
3. $= \frac{1}{12}$
4. Positive, so $\frac{5}{6}$ is the larger.

ANSWER

$\frac{1}{12}$

EXAMPLE 2 **Compute $4\frac{1}{5} - 1\frac{3}{5}$.**

1. Convert: $\frac{21}{5} - \frac{8}{5}$.

   Route A.

2. $= \frac{13}{5}$

3. $= 2\frac{3}{5}$

4. Check: $2\frac{3}{5} + 1\frac{3}{5} = 4\frac{1}{5}$ ✓

ANSWER

$2\frac{3}{5}$

EXAMPLE 3 **Find the distance between $-\frac{1}{2}$ and $\frac{1}{3}$ on the line.**

1. $-\frac{1}{2} - \frac{1}{3}$

2. $= -\frac{3}{6} - \frac{2}{6} = -\frac{5}{6}$

3. Take the modulus.

4. $\frac{5}{6}$

ANSWER

$\frac{5}{6}$

03 Interactive model

Ask which is larger, two thirds or three quarters, before any calculation; the class guesses, then subtracts, and the sign settles it.

Subtracting fractions

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
10	$2 \cdot 5$	2
4	2^2	5

LOWEST COMMON DENOMINATOR

$$20$$

$\frac{14}{20} - \frac{15}{20} = -\frac{1}{20}$ — the answer is negative.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Subtraction	Ayirish	Вычитание
Difference	Ayirma	Разность
Common denominator	Umumiy maxraj	Общий знаменатель
To borrow	Qarz olish	Занять
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Opposite	Qarama-qarshi	Противоположное
Simplify	Soddalashtirish	Упростить
Compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{2}{7} - \frac{5}{7}$ equals:

- $\frac{3}{7}$
- $-\frac{3}{7}$
- $\frac{7}{7}$
- $-\frac{7}{7}$

$\frac{1}{4} - (-\frac{1}{4})$ equals:

- 0
- $\frac{1}{2}$
- $-\frac{1}{2}$
- $\frac{1}{4}$

$\frac{1}{3} - \frac{1}{2}$ equals:

- $\frac{1}{6}$
- $-\frac{1}{6}$
- $\frac{1}{5}$
- $-\frac{1}{5}$

$4\frac{1}{5}$ with one whole borrowed:

- $3\frac{6}{5}$
- $3\frac{11}{5}$
- $3\frac{1}{5}$
- $5\frac{1}{5}$

The distance between $-\frac{1}{2}$ and $\frac{1}{3}$:

- $\frac{1}{6}$
- $\frac{5}{6}$
- $-\frac{5}{6}$
- $\frac{2}{5}$

A negative difference means:

- an error
- the second is larger
- the first is larger
- they are equal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{5}{7} - \frac{2}{7}$

ANSWER $\frac{3}{7}$

2. $\frac{2}{7} - \frac{5}{7}$

ANSWER $-\frac{3}{7}$

3. $\frac{3}{4} - \frac{1}{4}$

ANSWER $\frac{1}{2}$

4. $\frac{1}{4} - (-\frac{1}{4})$

ANSWER $\frac{1}{2}$

5. $\frac{1}{2} - \frac{1}{4}$

ANSWER $\frac{1}{4}$

6. $0.8 - 0.35$

ANSWER 0.45

7. $\frac{5}{8} - \frac{1}{8}$

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{5}{6} - \frac{3}{4}$

ANSWER $\frac{1}{12}$

2. $\frac{3}{4} - \frac{1}{6}$

ANSWER $\frac{7}{12}$

3. $\frac{1}{3} - \frac{1}{2}$

ANSWER $-\frac{1}{6}$

4. $-\frac{2}{5} - \frac{3}{10}$

ANSWER $-\frac{7}{10}$

5. $4\frac{1}{5} - 1\frac{3}{5}$

ANSWER $2\frac{3}{5}$

6. $3\frac{1}{4} - 1\frac{1}{2}$

ANSWER $1\frac{3}{4}$

7. Distance between $-\frac{1}{2}$ and $\frac{1}{3}$

ANSWER $\frac{5}{6}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{7}{8} - \frac{2}{3} - \frac{1}{6}$

ANSWER $\frac{1}{24}$

2. $-1\frac{1}{3} - (-2\frac{1}{2})$

ANSWER $1\frac{1}{6}$

3. $5 - 2\frac{3}{7}$

ANSWER $2\frac{4}{7}$

4. Find x: $\frac{3}{4} - x = \frac{1}{3}$

ANSWER $\frac{5}{12}$

5. Which is larger, $\frac{5}{8}$ or $\frac{7}{11}$?

ANSWER $\frac{7}{11}$

6. The difference between the largest and smallest of $\frac{1}{2}, -\frac{2}{3}, \frac{3}{4}$

ANSWER $\frac{17}{12}$

7. $0.6 - \frac{1}{4}$

ANSWER $\frac{7}{20}$

07 Homework

Homework — 5 tasks

Predict the sign of every answer before computing it.

1. Compute $\frac{7}{9} - \frac{2}{3}$ and $\frac{2}{5} - \frac{3}{4}$.
2. Compute $-\frac{1}{2} - \frac{2}{3}$.
3. Compute $5\frac{1}{4} - 2\frac{2}{3}$.
4. Find the distance between $-\frac{3}{4}$ and $\frac{1}{2}$.
5. Find x if $\frac{2}{3} - x = \frac{1}{6}$.

Addition and subtraction of rational numbers together

Long chains of fractions — one common denominator, one pass.

LESSON 43 | 1 × 40 MIN | MATEMATIKA 7, §18 | S8 7.2

LESSON CLOCK

10' | 12' | 12' | 6'

One denominator for all	10 min
Mixing forms	12 min
Simple equations	12 min
Homework	6 min

LEARNING OBJECTIVES

- Evaluate a chain of additions and subtractions of rational numbers.
- Use one common denominator for the whole chain.
- Combine fractions and decimals in one calculation.
- Solve a simple equation of the form x + a = b in $\mathbb{Q}$.

TEXTBOOK

National	pp. 114–117
Cambridge	Stage 8, pp. 70–76
Workbook	Exercise 7.2

01 Explanation

One denominator for all

For a chain such as $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$, find the LCM of **all** the denominators at once and rewrite everything.

$$\frac{6}{12} - \frac{8}{12} + \frac{9}{12} = \frac{7}{12}$$

CHAIN	LCM	ANSWER
$\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$	*12*	$\frac{7}{12}$
$\frac{2}{5} + \frac{1}{2} - \frac{7}{10}$	*10*	$\frac{1}{5}$
$-\frac{1}{3} - \frac{1}{4} + \frac{5}{6}$	*12*	$\frac{1}{4}$

ONE CONVERSION, NOT TWO

Working pairwise means converting twice and simplifying twice. Converting the whole chain once is shorter and far less error-prone.

Mixing forms

When fractions and decimals appear together, convert to whichever form is easier.

EXPRESSION	BEST FORM	ANSWER
$0.5 + \frac{1}{4}$	fractions	$\frac{3}{4}$
$0.25 - \frac{1}{3}$	fractions	$-\frac{1}{12}$
$\frac{1}{2} + 0.3$	decimals	0.8
$\frac{1}{3} + 0.2$	fractions	$\frac{8}{15}$

USE FRACTIONS WHENEVER A THIRD IS INVOLVED

$\frac{1}{3}$ has no exact decimal form, so any decimal working is an approximation. Fractions keep the answer exact.

Simple equations

$x + a = b$ is solved by subtracting a from both sides:

$$x = b - a$$

EQUATION	WORKING	ANSWER
$x + \frac{1}{3} = \frac{5}{6}$	$\frac{5}{6} - \frac{2}{6}$	$\frac{1}{2}$
$x - \frac{2}{5} = \frac{1}{10}$	$\frac{1}{10} + \frac{4}{10}$	$\frac{1}{2}$
$x + \frac{3}{4} = \frac{1}{4}$	$\frac{1}{4} - \frac{3}{4}$	$-\frac{1}{2}$

ALWAYS SUBSTITUTE THE ANSWER BACK

One line of checking catches every sign error and every denominator slip, and the check is quicker than the solution.

02 Worked examples

EXAMPLE 1 **Compute $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$.**

1. LCM of $2, 3, 4$ is 12.
2. $$\frac{6}{12} - \frac{8}{12} + \frac{9}{12}$$
3. $$= \frac{6 - 8 + 9}{12}$$
4. $$= \frac{7}{12}$$

ANSWER

$$\frac{7}{12}$$

EXAMPLE 2 **Compute $0.25 - \frac{1}{3}$.**

1. $0.25 = \frac{1}{4}$

   Fractions are exact.

2. LCM of 4 and 3 is 12.

3. $\frac{3}{12} - \frac{4}{12}$

4. $= -\frac{1}{12}$

ANSWER

$-\frac{1}{12}$

EXAMPLE 3 **Solve $x + \frac{3}{4} = \frac{1}{4}$.**

1. $x = \frac{1}{4} - \frac{3}{4}$

2. $= -\frac{2}{4}$

3. $= -\frac{1}{2}$

4. Check: $-\frac{1}{2} + \frac{3}{4} = \frac{1}{4}$ ✓

ANSWER

$x = -\frac{1}{2}$

03 Interactive model

Do one chain twice on the board — pairwise and all at once — and let the class time both; the single conversion wins visibly.

One denominator for the whole chain

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
2	2	6
3	3	4
4	2^2	3

LOWEST COMMON DENOMINATOR

$$12$$

$$\frac{6}{12} + \frac{8}{12} - \frac{9}{12} = \frac{5}{12}.$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Chain of operations	Amallar zanjiri	Цепочка действий
Common denominator	Umumiy maxraj	Общий знаменатель
To combine	Birlashtirish	Объединить
Equation	Tenglama	Уравнение
Unknown	Noma’lum	Неизвестное
Check	Tekshirish	Проверка
Decimal form	O‘nli ko‘rinish	Десятичная форма
Simplest form	Eng sodda ko‘rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For a chain, use the LCM of:

- the first two
- all the denominators
- the largest
- the smallest

$\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$ equals:

- $\frac{7}{12}$
- $\frac{5}{12}$
- $\frac{11}{12}$
- $-\frac{7}{12}$

$0.25 - \frac{1}{3}$ equals:

- $\frac{1}{12}$
- $-\frac{1}{12}$
- -0.08
- $\frac{7}{12}$

When a third is involved, prefer:

- decimals
- fractions
- either
- percentages

$x + \frac{3}{4} = \frac{1}{4}$ gives:

- 1
- $\frac{1}{2}$
- $-\frac{1}{2}$
- -1

After solving you should:

- stop
- substitute back
- round
- draw a graph

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{1}{2} + \frac{1}{3}$

ANSWER $\frac{5}{6}$

2. $\frac{1}{2} - \frac{1}{3}$

ANSWER $\frac{1}{6}$

3. $0.5 + \frac{1}{4}$

ANSWER $\frac{3}{4}$

4. $\frac{1}{2} + 0.3$

ANSWER 0.8

5. Solve $x + \frac{1}{3} = \frac{5}{6}$

ANSWER $\frac{1}{2}$

6. Solve $x - \frac{2}{5} = \frac{1}{10}$

ANSWER $\frac{1}{2}$

7. $\frac{3}{8} + \frac{1}{8} - \frac{1}{4}$

ANSWER $\frac{1}{4}$

Medium · the standard question

7 PROBLEMS

1. $\frac{1}{2} - \frac{2}{3} + \frac{3}{4}$

ANSWER $\frac{7}{12}$

2. $\frac{2}{5} + \frac{1}{2} - \frac{7}{10}$

ANSWER $\frac{1}{5}$

3. $-\frac{1}{3} - \frac{1}{4} + \frac{5}{6}$

ANSWER $\frac{1}{4}$

4. $0.25 - \frac{1}{3}$

ANSWER $-\frac{1}{12}$

5. $\frac{1}{3} + 0.2$

ANSWER $\frac{8}{15}$

6. Solve $x + \frac{3}{4} = \frac{1}{4}$

ANSWER $-\frac{1}{2}$

7. $1 - \frac{1}{2} - \frac{1}{3}$

ANSWER $\frac{1}{6}$

Hard · stretch and reason

7 PROBLEMS

1. $\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{6}$

ANSWER $\frac{1}{4}$

2. $-\frac{3}{5} + 0.25 - \frac{1}{4}$

ANSWER $-\frac{3}{5}$

3. $2\frac{1}{3} - 1\frac{1}{2} + \frac{5}{6}$

ANSWER $1\frac{2}{3}$

4. Solve $\frac{2}{3} - x = -\frac{1}{6}$

ANSWER $\frac{5}{6}$

5. Three pieces of a rod are $\frac{1}{4}, \frac{1}{3}, \frac{1}{6}$ of it: the rest

ANSWER $\frac{1}{4}$

6. $0.75 - \frac{2}{3} + \frac{1}{12}$

ANSWER $\frac{1}{6}$

7. The mean of $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$

ANSWER $\frac{1}{3}$

07 Homework

Homework — 5 tasks

Use one common denominator for each whole chain, and check every equation.

1. Compute $\frac{2}{3} - \frac{1}{2} + \frac{1}{6}$.
2. Compute $-\frac{1}{4} + \frac{2}{5} - \frac{1}{10}$.
3. Compute $0.4 + \frac{1}{3}$.
4. Solve $x + \frac{5}{6} = \frac{1}{3}$.
5. Solve $\frac{3}{4} - x = \frac{1}{8}$.

Control work 3, and work on the mistakes

Rational numbers in one paper, and the first quarter closed.

LESSON 44–45 | 2 × 40 MIN | MATEMATIKA 7, NAZORAT ISHI 3 | S8 7 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The quarter	5 min

LEARNING OBJECTIVES

- Convert between fractions, decimals and mixed numbers under time.
- Add and subtract rational numbers with signs.
- Solve a simple equation in $\mathbb{Q}$.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 95–117
Cambridge	Stage 8, pp. 66–76
Workbook	Control paper M3

01 Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Write -2.4 and $3\frac{1}{4}$ as fractions $\frac{p}{q}$	*4*	L37–38
2	Convert $\frac{7}{8}$ and $\frac{5}{12}$ to decimals	*4*	L37–38
3	Compute $\frac{2}{3}+\frac{3}{4}$ and $-\frac{1}{2}+\frac{1}{5}$	*6*	L39–40
4	Compute $\frac{5}{6}-\frac{3}{8}$ and $-\frac{2}{5}-\frac{1}{2}$	*6*	L41–42
5	Compute $3\frac{1}{3}-1\frac{3}{4}$	*5*	L41–42
6	Solve $x+\frac{2}{3}=\frac{1}{6}$	*5*	L43

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for simplifying; Q2 one for showing the recurrence; Q3 two for the LCM; Q5 two for converting to improper fractions; Q6 one for the check.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
fraction not simplified	$-\frac{24}{10}$	$-\frac{12}{5}$
recurrence not shown	0.41	$0.41666\ldots$
denominators added	$\frac{2}{3}+\frac{3}{4}=\frac{5}{7}$	$\frac{17}{12}$
wrong sign kept	$-\frac{1}{2}+\frac{1}{5}=\frac{3}{10}$	$-\frac{3}{10}$
mixed number sign	$-2\frac{1}{3}=-2+\frac{1}{3}$	$-\frac{7}{3}$
borrowing wrong	$3\frac{1}{3}=2\frac{11}{3}$	$2\frac{4}{3}$
equation solved by adding	$x=\frac{1}{6}+\frac{2}{3}$	$x=\frac{1}{6}-\frac{2}{3}$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The quarter in one page

BLOCK	THE SENTENCE
the basic figures	point, line and plane, and the axioms that govern them
segments and angles	both add when a point or a ray lies inside
perpendicular lines	adjacent add to $180°$, vertical are equal
the integers	invented so that subtraction always works
the sign rules	alike gives $+$, unlike gives $-$
the rationals	invented so that division always works
fractions	a common denominator, then the numerators

LOOKING FORWARD

Quarter II returns to geometry for the congruence of triangles, then opens algebra proper with powers, monomials and polynomials. The sign rules of this quarter are used in every line of it.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: $\frac{2}{3} + \frac{3}{4}$ and $-\frac{1}{2} + \frac{1}{5}$.**

1. LCM of *3* and *4* is *12*.

2. $\frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1\frac{5}{12}$

3. LCM of *2* and *5* is *10*.

4. $-\frac{5}{10} + \frac{2}{10} = -\frac{3}{10}$

   The half is bigger.

ANSWER

$1\frac{5}{12}$ and $-\frac{3}{10}$

EXAMPLE 2 **Model answer, Q5: $3\frac{1}{3} - 1\frac{3}{4}$.**

1. Convert: $\frac{10}{3} - \frac{7}{4}$.

2. LCM 12: $\frac{40}{12} - \frac{21}{12}$.

3. $= \frac{19}{12}$

4. $= 1\frac{7}{12}$

ANSWER

$1\frac{7}{12}$

EXAMPLE 3 **Model answer, Q6: solve $x + \frac{2}{3} = \frac{1}{6}$.**

1. $x = \frac{1}{6} - \frac{2}{3}$

   Subtract, do not add.

2. $= \frac{1}{6} - \frac{4}{6}$

3. $= -\frac{3}{6} = -\frac{1}{2}$

4. Check: $-\frac{1}{2} + \frac{2}{3} = \frac{1}{6}$ ✓

ANSWER

$x = -\frac{1}{2}$

03 Interactive model

Mark Q3 aloud with a paper strip folded into twelfths; the wrong answer of five sevenths becomes visibly impossible.

Chapter III in twelve questions

A rational number is:

- any decimal
- $\frac{p}{q}$ with $q \neq 0$
- an integer
- a fraction below 1

Is -4 rational?

- yes
- no
- sometimes
- undefined

$\frac{3}{8}$ as a decimal:

- 0.38
- 0.375
- 0.83
- 0.125

A fraction terminates when the denominator has only:

- 2 and 3
- 2 and 5
- primes
- odd factors

$\frac{2}{7} + \frac{3}{7}$ equals:

- $\frac{5}{14}$
- $\frac{5}{7}$
- $\frac{6}{7}$
- $\frac{6}{49}$

For unlike denominators, use:

- the product
- the LCM
- the HCF
- the sum

$-\frac{1}{2} + \frac{1}{3}$ equals:

- $\frac{1}{6}$
- $-\frac{1}{6}$
- $\frac{5}{6}$
- $-\frac{5}{6}$

$a - b$ equals:

- $a + b$
- $a + (-b)$
- $b - a$
- $-a + b$

$-2\frac{1}{3}$ as an improper fraction:

- $-\frac{7}{3}$
- $-\frac{5}{3}$
- $\frac{7}{3}$
- $-\frac{1}{3}$

$4\frac{1}{5}$ with one whole borrowed:

- $3\frac{6}{5}$
- $3\frac{11}{5}$
- $3\frac{1}{5}$
- $5\frac{1}{5}$

For a chain, take the LCM of:

- the first two
- all of them
- the largest
- the smallest

$x + a = b$ gives:

- $x = a + b$
- $x = b - a$
- $x = a - b$
- $x = ab$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Rational number	Ratsional son	Рациональное число
Common denominator	Umumiy maxraj	Общий знаменатель
Mixed number	Aralash son	Смешанное число
Recurring decimal	Davriy o‘nli kasr	Периодическая дробь
Simplest form	Eng sodda ko‘rinish	Несократимый вид
Equation	Tenglama	Уравнение
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 loses a mark most often for:

- a wrong sign
- not simplifying
- a decimal
- no units

Q2 must show:

- a fraction
- the recurrence
- a percentage
- rounding

Q3 begins with:

- the numerators
- the LCM
- the answer
- a sketch

In Q5, converting to improper fractions avoids:

- the LCM
- borrowing
- signs
- checking

Q6 is solved by:

- adding $\frac{2}{3}$
- subtracting $\frac{2}{3}$
- multiplying
- dividing

Work on the mistakes means:

- fix the wrong line
- rewrite the solution
- copy the answer
- skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. -2.4 as a fraction

ANSWER $-\dfrac{12}{5}$

2. $3\dfrac{1}{4}$ as a fraction

ANSWER $\dfrac{13}{4}$

3. $\dfrac{7}{8}$ as a decimal

ANSWER 0.875

4. $\dfrac{5}{12}$ as a decimal

ANSWER $0.41666\ldots$

5. $\dfrac{2}{3}+\dfrac{3}{4}$

ANSWER $1\dfrac{5}{12}$

6. $-\dfrac{1}{2}+\dfrac{1}{5}$

ANSWER $-\dfrac{3}{10}$

7. $\dfrac{5}{6}-\dfrac{3}{8}$

ANSWER $\dfrac{11}{24}$

Medium · the standard question

7 PROBLEMS

1. $-\dfrac{2}{5} - \dfrac{1}{2}$

ANSWER $-\dfrac{9}{10}$

2. $3\dfrac{1}{3} - 1\dfrac{3}{4}$

ANSWER $1\dfrac{7}{12}$

3. Solve $x + \dfrac{2}{3} = \dfrac{1}{6}$

ANSWER $-\dfrac{1}{2}$

4. $\dfrac{1}{2} + \dfrac{1}{3} - \dfrac{1}{4}$

ANSWER $\dfrac{7}{12}$

5. $0.6 + \dfrac{1}{5}$

ANSWER $\dfrac{4}{5}$

6. Simplify $\dfrac{45}{75}$

ANSWER $\dfrac{3}{5}$

7. Solve $x - \dfrac{1}{4} = \dfrac{1}{2}$

ANSWER $\dfrac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{1}{6} - 3\frac{1}{2}$

ANSWER $-1\frac{1}{3}$

2. $-\frac{3}{4} + \frac{5}{6} - \frac{1}{12}$

ANSWER 0

3. Which is larger, $\frac{7}{12}$ or 0.6?

ANSWER 0.6

4. Solve $\frac{5}{6} - x = -\frac{1}{3}$

ANSWER $\frac{7}{6}$

5. The mean of $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$

ANSWER $\frac{1}{2}$

6. A tank $\frac{2}{3}$ full loses $\frac{1}{4}$ of the tank

ANSWER $\frac{5}{12}$ full

7. Order $\frac{3}{5}, \frac{5}{8}, \frac{7}{12}$

ANSWER $\frac{7}{12} < \frac{3}{5} < \frac{5}{8}$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the second quarter begins.

1. Write -1.8 and $2\frac{3}{5}$ as fractions.
2. Convert $\frac{9}{16}$ and $\frac{7}{15}$ to decimals.
3. Compute $\frac{3}{5} + \frac{1}{4}$ and $-\frac{2}{3} + \frac{1}{6}$.
4. Compute $4\frac{1}{4} - 2\frac{2}{3}$.
5. Solve $x + \frac{3}{5} = \frac{1}{10}$.